# Information Retention in Phase-Saturated Objects

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**Abstract**

We demonstrate, within the framework of Phase Theory, that information is fully retained throughout the complete lifecycle of a phase-saturated object — the Phase Theory analog of a gravitational black hole. Phase Theory posits that physical reality consists of a single phase-bearing substrate $\Phi : M \to T$, where M is a pre-geometric topological base space and T is a compact Lie group, with all spacetime structure, matter, and geometry emerging from the global organization of this substrate. Within this framework, black holes correspond to regions in which the phase-inconsistency functional $I[\Phi]$ saturates or exceeds its admissibility threshold, yielding what we term phase-saturated objects. We show that the black hole information paradox, as classically formulated, rests on three mutually reinforcing errors: (1) the identification of spacetime geometry as a fundamental rather than emergent structure, (2) the treatment of information as a locally encoded property of degrees of freedom tied to spacetime points, and (3) the misinterpretation of the thermal character of Hawking radiation as evidence of genuine randomness or state destruction. Phase Theory corrects all three errors simultaneously. We prove the Phase-Topology Persistence Theorem (Theorem 1), which establishes that all homotopy-class invariants of Φ are preserved continuously through phase-saturation events. We prove the Information Retention Theorem (Theorem 2), establishing that the phase-information content $I_{phase}(\Phi_0)$ of any initial configuration equals that of the final post-evaporation configuration $\Phi_\infty$. Hawking radiation is reinterpreted as admissibility leakage: outgoing phase modes that carry global phase consistency away from the saturated region without severing correlations with the interior state. The apparent thermal character of this radiation is a projection artifact of geometric coarse-graining. The Page curve is recovered structurally from phase-saturation dynamics without recourse to replica wormholes, island formulae, non-perturbative gravitational saddle points, or horizon microstates. The AMPS firewall does not arise because Phase Theory admits no localized quantum states at the horizon, no monogamy condition on phase correlations, and no physical membrane at the saturation boundary. Black hole complementarity is unnecessary because spacetime is not a primary structure in Phase Theory. Six concrete observational predictions distinguishing Phase Theory from semiclassical gravity are presented. We situate this work as Paper 50 in the Phase Theory series and identify open questions for subsequent investigations.

## 1. Why the Information Paradox Exists

The black hole information paradox stands as one of the most productive conceptual crises in theoretical physics, having driven the development of holography, the anti-de Sitter/conformal field theory (AdS/CFT) correspondence, the island formula, the Page curve derivation via replica wormholes, and a host of related developments. It is a paradox of remarkable durability: proposed in its essential form by Hawking in his landmark 1974 paper on particle creation by black holes [1] and sharpened in his 1976 analysis of information loss [2], it has resisted decisive resolution for over five decades. Understanding precisely why the paradox is so durable requires a careful analysis of its logical structure, since the paradox is not a single proposition but rather the tension among three independent claims, each individually well-supported, which collectively cannot all be true.

The first claim is that spacetime geometry is a fundamental physical structure. General relativity, as formulated by Einstein and subsequently developed into a precise mathematical framework, treats the metric tensor $g_{\mu\nu}$ as a dynamical field on a smooth Lorentzian manifold. The black hole solutions of general relativity — Schwarzschild, Reissner-Nordström, Kerr, Kerr-Newman — describe regions of spacetime bounded by event horizons: one-way causal membranes through which information, in the geometric sense, cannot pass outward. Penrose diagrams, conformal compactifications of spacetime, give these solutions a precise global causal structure. In the Penrose diagram of an evaporating black hole, the spacelike singularity and the region interior to the horizon are causally disconnected from future null infinity, which is where an asymptotic observer would receive the Hawking radiation. This causal structure is a direct consequence of treating $g_{\mu\nu}$ as fundamental. The information trapped inside the event horizon appears causally incapable of reaching any outside observer.

The second claim is that quantum evolution is unitary: the time evolution of a quantum state $|\psi(t)\rangle$ is governed by a unitary operator $U(t) = \exp(-iHt/\hbar)$, which is invertible, and which preserves the von Neumann entropy $S = -\mathrm{Tr}(\rho \log \rho)$ of a pure state. If the universe begins in a pure state, it must end in a pure state. This is not an optional feature of quantum mechanics; it is constitutive of its probabilistic interpretation. Preskill's 1992 lectures [3] set the stage for the modern formulation of the paradox by making this tension precise: if Hawking radiation truly carries no information about the initial state, then the process of black hole formation and evaporation maps a pure state to a mixed thermal state, a manifestly non-unitary evolution. The paradox is that this non-unitary evolution seems to be an unavoidable prediction of combining general relativity with quantum field theory.

The third claim is that Hawking radiation is thermal and memoryless. Hawking's original derivation [1, 2] proceeded by considering a quantum field in the fixed background geometry of a collapsing star, tracing the Bogoliubov transformation relating the vacuum state defined by early-time inertial observers to the vacuum state defined by late-time observers at future null infinity. The result is that the late-time observer detects a thermal spectrum of particles at the Hawking temperature $T_H = \hbar c^3/(8\pi G k_B M)$, which depends only on the black hole mass M. The radiation has no memory of how the black hole was formed — it depends only on M, Q, and J (mass, charge, and angular momentum). Bekenstein had already shown in 1973 [4] that the entropy of a black hole is proportional to its horizon area: $S_{BH} = A/(4G\hbar)$, and Hawking's thermal emission is consistent with this entropy decreasing as the black hole evaporates. The thermal, state-independent character of the radiation is what makes it appear information-destroying.

These three claims are mutually inconsistent. If geometry is fundamental and defines a causal horizon, if quantum evolution is unitary, and if Hawking radiation is genuinely thermal, then information both must and must not be preserved — must, because of unitarity; must not, because the causal structure seals the interior from future null infinity and the radiation carries no memory. This is the paradox in its cleanest form. Various attempts at resolution over the decades have tried to modify or reject one of the three claims without abandoning the others. Remnant proposals [5] preserve information in a stable sub-Planckian relic at the end of evaporation — but this requires infinite internal states, contradicting the Bekenstein bound. Black hole complementarity (Susskind, Thorlacius, and Uglum, 1993 [6]) attempts to reconcile the infalling and external observer descriptions by declaring them mutually exclusive — but this is conceptually unstable and leads to its own paradoxes. The AMPS firewall [7] (Almheiri, Marolf, Polchinski, Sully, 2013) argues that unitarity and the monogamy of entanglement together imply a Planck-scale energy density at the horizon, violating the equivalence principle. Holography and AdS/CFT [8] (Maldacena, 1997) offer the most technically sophisticated response, but require a bulk-boundary duality whose ontological status remains contested. The island formula [9, 10] (Penington 2019; Almheiri et al. 2019) recovers the Page curve via non-perturbative gravitational saddle points, but the physical interpretation of replica wormholes remains under active debate.

Phase Theory takes a fundamentally different approach. Rather than modifying one of the three claims while preserving the others, it observes that the first claim — the fundamentality of spacetime geometry — is an assumption that Phase Theory explicitly rejects at the axiomatic level. Spacetime geometry, in Phase Theory, is an emergent description valid only in the dilute-phase regime. At phase-saturated densities, this description breaks down entirely. The paradox, in Phase Theory, is therefore dissolved rather than resolved: it does not arise as a genuine tension, because the geometric framework in which it is formulated is not applicable to the regime in which the alleged information loss occurs. The information was never in the geometry. It was, from the beginning, in the global phase structure — which is perfectly well-defined throughout the entire lifecycle of a phase-saturated object, including formation, evolution, and evaporation.

## 2. Foundations of Phase Theory: A Concise Review

Phase Theory is a foundational framework for physics in which the sole primitive constituent of physical reality is a phase-bearing substrate. We review here the five axioms and the central mathematical structures relevant to the present work; for a more complete treatment, the reader is referred to Papers 1 through 11 of this series, and in particular to Paper 1 for the foundational ontology, Paper 3 for the topological structure of the configuration space, and Paper 12 for the treatment of information as phase distinguishability.

The fundamental object of Phase Theory is a map $\Phi : M \to T$, where M is a topological space (compact or with appropriate boundary conditions) that carries no pre-specified metric structure, and T is a compact Lie group. The space M is the base topological space — a pre-geometric substratum on which no notion of distance, angle, or duration is defined a priori. The group T encodes the phase degrees of freedom. The specific choice of T is constrained by the axioms but not uniquely determined by them; different admissible choices of T correspond to different matter content in the emergent geometric description. The map Φ is subject to regularity conditions appropriate to the topology of M and T, and the space of all such maps forms the configuration space $\Gamma = \{\Phi : M \to T \mid I[\Phi] < \infty\}$, where $I[\Phi]$ is the phase-inconsistency functional defined in Axiom 1.

**Axiom 1 (Global Phase Consistency).** Physical configurations are those that minimize the phase-inconsistency functional

$$I[\Phi] = \int_M \rho(\Phi, D\Phi)\, d\mu, \quad (1)$$

where $\rho = \frac{1}{2} K_{ab}(\Phi)\, D_\mu\Phi^a\, D^\mu\Phi^b + V(\Phi) + \lambda\, \mathrm{Tr}(F_{\mu\nu}\, F^{\mu\nu})$, and the minimization is performed subject to topological constraints. Here $K_{ab}$ is a positive-definite metric on the target Lie algebra, $D_\mu$ is the covariant derivative with respect to a connection on T, $V(\Phi)$ is a phase potential encoding symmetry-breaking structure, λ is a coupling constant, and $F_{\mu\nu}$ is the curvature of the connection. The measure dμ is a topologically invariant measure on M (not derived from any metric). The functional $I[\Phi]$ measures the degree of phase inconsistency in a configuration: small I means high internal phase coherence; large I means high internal phase tension or disorder.

**Axiom 2 (Update Ordering).** The evolution of the phase configuration Φ is governed by a partial order on updates, realized as a directed acyclic graph (DAG). Time in Phase Theory is not a fundamental parameter but an emergent ordering of admissible phase updates. The acyclicity of the DAG prevents causal loops. The emergent time coordinate arises from the linearization of this partial order in the dilute-phase regime, where it converges to the time coordinate of standard relativistic field theory.

**Axiom 3 (Topological Stability).** Topological defects in Φ are classified by the homotopy groups $\pi_n(T)$, by holonomy classes, linking numbers, and knot invariants of the defect loci. Each homotopy class corresponds to a stable particle species in the emergent geometric description. The stability of defects follows from the topological character of the classification: continuous deformations of Φ cannot change its homotopy class, and therefore particle number (in the appropriate topological sense) is conserved.

**Axiom 4 (Coherence Limitation).** The mutual phase information between any two regions A and B of M satisfies the bound

$$J(\Phi_A; \Phi_B) \leq C \cdot \min(|\partial A|, |\partial B|), \quad (2)$$

where $J(\Phi_A; \Phi_B)$ is the mutual phase information between the restrictions of Φ to A and B, $|\partial A|$ denotes the topological measure of the boundary of A (which reduces to area in the emergent geometric description), and C is a constant with dimensions of inverse length squared in the emergent description. This axiom is the Phase Theory analog of the Bekenstein bound, and it implies that information content is bounded by boundary area rather than volume — a topological version of the holographic principle that does not require a holographic screen to be fundamental.

**Axiom 5 (Metric Emergence).** The emergent spacetime metric is given by

$$g_{\mu\nu}(x) = \alpha \langle \nabla_\mu \Phi(x) \cdot \nabla_\nu \Phi(x) \rangle_{coh}, \quad (3)$$

where $\alpha$ is a constant with appropriate dimensions (set by the emergent Newton constant G), and the expectation value is taken over a coherence domain — a region of M over which the phase field is approximately uniform at the microscopic scale. The metric is a derived quantity, not a fundamental one. It is defined only where coherence domains exist. Where coherence domains collapse to the Planck scale or below, the geometric description breaks down, and $g_{\mu\nu}$ ceases to be well-defined. This occurs precisely in the saturated regime.

The admissibility constraint system is the collection of all conditions that a physical configuration must satisfy: finite phase-inconsistency ($I[\Phi] < \infty$), topological consistency (defect loci are smooth submanifolds with well-defined homotopy class), and coherence limitation (Axiom 4). A configuration satisfying all admissibility constraints is called admissible. The space of admissible configurations is a proper subspace of $\Gamma$, and it is stratified by topological sector. Within each topological sector, the admissible configurations form a connected component, and the evolution governed by Axiom 1 keeps the system within this connected component.

A phase-saturated regime is one in which $I[\Phi|_\Omega] \geq I_{sat}$ for some compact subregion $\Omega \subset M$, where $I_{sat}$ is the admissibility saturation threshold. This threshold is determined by the topological density of defects that the substrate can accommodate without losing global coherence. In a phase-saturated region, coherence domains shrink to Planck scale, the metric emergent from Axiom 5 diverges or becomes singular, and the geometric description of the region is no longer valid. This is the Phase Theory analog of a black hole interior. The boundary of the saturated region — where $I[\Phi] = I_{sat}$ — is the analog of the event horizon, and as we show in Section 8, it corresponds precisely to the locus where $g_{\mu\nu}$ would become singular in the geometric description.

## 3. What Information Is in Phase Theory

A precise definition of information is indispensable for any discussion of the information paradox. In standard quantum information theory, information is identified with von Neumann entropy $S(\rho) = -\mathrm{Tr}(\rho \log \rho)$ of a density matrix $\rho$, with Shannon entropy $H(X) = -\sum_i p_i \log p_i$ of a classical probability distribution, or with the mutual information $I(A:B) = S(A) + S(B) - S(AB)$ between subsystems. All of these notions presuppose a Hilbert space, a factorization of that Hilbert space into subsystems, and a notion of local degrees of freedom. In the holographic context, information is sometimes identified with the area of the Ryu-Takayanagi surface [11] in a bulk spacetime, which measures the entanglement entropy of a boundary region. All of these definitions share a common feature: they are tied, in different ways, to a geometric or combinatorial substrate that Phase Theory treats as emergent.

Phase Theory requires a definition of information that is native to the phase-field description and that does not presuppose geometric structure. We define information as *phase distinguishability*: the persistence of globally admissible distinctions among phase configurations. More precisely, two configurations Φ and Φ' carry distinguishable information if and only if they lie in different connected components of the admissible configuration space $\Gamma_{adm}$, or if they are connected by a path in $\Gamma_{adm}$ that passes through configurations with different topological invariants. The information content of a configuration is not a property of a single point in Γ but a property of the connected component (topological sector) to which the configuration belongs.

Formally, we define the phase-information content of a configuration Φ as

$$I_{phase}(\Phi) = \dim\{\delta\Phi : \delta I[\Phi] \geq 0, \text{admissible}\}, \quad (4)$$

where the dimension is taken in the sense of the dimension of the space of admissible deformations that are distinguishable from Φ at the phase level. This is the dimensionality of the space of admissible phase-consistent distinctions around Φ. Intuitively, $I_{phase}(\Phi)$ counts the number of independent ways in which the configuration can be varied while remaining admissible and while remaining distinguishable from Φ by some admissibility-preserving observable. This is a genuinely global quantity: it is not the sum of local degrees of freedom at individual points of M, but a property of the global configuration space structure.

The contrast with von Neumann entropy is instructive. Von Neumann entropy S(ρ) measures uncertainty about which pure state a system is in, given that we know only the density matrix ρ. It is a property of a reduced description — a description obtained by tracing out degrees of freedom associated with some spatial region. $I_{phase}(\Phi)$, by contrast, is a property of the full global configuration. It does not decrease when we trace out a subregion, because it is not defined in terms of a partial trace. When we restrict $\Phi$ to a subregion and lose information about the complement, we are not destroying phase information — we are adopting an inadequate description of a configuration whose information content is intrinsically global.

The contrast with Shannon entropy is equally sharp. Shannon entropy $H(X)$ is defined for a classical random variable $X$ with a probability distribution over a finite alphabet. It measures ignorance about the value of $X$, given a probability distribution. Phase information $I_{phase}(\Phi)$ is not a measure of ignorance about anything: it is a measure of the structural richness of the configuration space around $\Phi$. A configuration with high phase information is one that can be distinguished from many other admissible configurations; one with low phase information is one that is nearly uniquely determined by the topological invariants alone. This structural richness is preserved by continuous evolution in the topological sense, because continuous evolution cannot change the number of topological sectors accessible from a given configuration.

In the holographic context, it is tempting to identify information with the area of a minimal surface in the bulk, as in the Ryu-Takayanagi formula [11]. Phase Theory does not dispute that the Ryu-Takayanagi formula gives the correct answer for entanglement entropy in the dilute-phase (geometric) regime; what it disputes is the claim that this geometric formula captures a fundamental property. The phase-level version of the holographic bound is Axiom 4 (Coherence Limitation), which bounds the mutual phase information $J(\Phi_A; \Phi_B)$ by the topological measure of the boundary $|\partial A|$. The Ryu-Takayanagi formula emerges as the geometric realization of this bound in the dilute regime, where areas and boundary measures coincide. In the saturated regime, the Ryu-Takayanagi formula is inapplicable because the geometric description fails, but Axiom 4 continues to hold at the phase-native level.

## 4. Phase-Saturated Objects: Definition and Structure

We now provide a precise definition of phase-saturated objects and describe their internal structure in detail. A phase-saturated object is a compact subregion $\Omega \subset M$ together with a phase configuration $\Phi|_\Omega$ such that the phase-inconsistency functional satisfies $I[\Phi|_\Omega] \geq I_{sat}$, where $I_{sat}$ is the admissibility saturation threshold. The saturation threshold $I_{sat}$ is the value of I at which the coherence domain size $L_{coh}(x)$ — defined as the correlation length of the phase field at the point x — reaches the Planck length $L_{Pl}$. Below $I_{sat}$, coherence domains are macroscopic and the geometric description (Axiom 5) is valid. At $I_{sat}$, coherence domains collapse to the Planck scale and geometric descriptions fail.

The internal structure of a phase-saturated object is characterized by several features that distinguish it sharply from the geometric description of a black hole interior. First, the topological defect density within Ω has reached its maximum admissible value. Defects — topological structures classified by $\pi_n(T)$ — are packed as densely as the admissibility constraints permit. Their linking numbers, holonomies, and winding numbers are well-defined at the phase level even though the geometric description of their locations (as points in spacetime) has broken down. Second, the phase gradient $|D\Phi|$ is everywhere at its maximum admissible value, consistent with the boundary condition $I[\Phi]|_{\partial\Omega sat} = I_{sat}$. Third, the coherence term in $g_{\mu\nu}$ — the averaged product $\langle \nabla_\mu\Phi \cdot \nabla_\nu\Phi \rangle_{coh}$ — is dominated by short-wavelength fluctuations and ceases to define a smooth tensor field. The emergent metric becomes singular, and the interior of Ω is not geometrically accessible.

The analogy with a classical black hole is precise in the semiclassical limit. When the phase-inconsistency density is small (dilute phase regime), the emergent metric $g_{\mu\nu}$ satisfies the Einstein equations with a source determined by the phase configuration (this is derived in Paper 27 of this series). The Schwarzschild solution emerges for a spherically symmetric concentration of topological defects with total winding number k in the dilute regime. As the defect density increases — as the object becomes more massive in the geometric description — the local phase-inconsistency increases. When $I[\Phi|_\Omega]$ reaches $I_{sat}$, the system enters the phase-saturated regime and the Schwarzschild description breaks down. The Schwarzschild horizon radius

$r_s = 2GM/c^2$ corresponds, in Phase Theory, to the radius of the saturation boundary $\partial\Omega_{sat}$ in the emergent geometric description.

The interior of the phase-saturated object is not empty in any meaningful sense. It is phase-dense: it contains the highest topological defect density that the admissibility constraints permit, it carries all the topological invariants of the infallen matter, and it is connected to the exterior through the global phase correlation function G(x, y) (Section 13). What the interior lacks is a geometric description: there is no smooth metric on $\Omega$, no well-defined notion of spatial distance, and no notion of temporal ordering that reduces to the usual Lorentzian time in that region. This is not a physical absence of structure but a failure of the emergent geometric description to apply. The phase field $\Phi$ is well-defined and globally smooth throughout $\Omega \cup \partial\Omega_{sat} \cup (M \setminus \Omega)$ — there is no singularity in the phase field itself, only in its geometric projection.

## 5. The Collapse Process in Phase-Native Language

Gravitational collapse, in Phase Theory, is a process by which a collection of topological defects — the phase-field analogs of massive particles — concentrates in a compact region of M, driving the local phase-inconsistency functional toward and eventually past the saturation threshold. We describe this process in detail using the phase-native language, without recourse to emergent geometric quantities except where they illuminate the connection to standard results.

Consider an initial configuration $\Phi_0$ representing a distribution of topological defects with total winding number $K_{total}$, linking numbers $L_{ij}$, and holonomies $H_i$. In the geometric description, this corresponds to a gas or cloud of matter particles with total mass M, charge Q, and angular momentum J. The initial phase-inconsistency $I[\Phi_0|_{\Omega 0}]$ is well below $I_{sat}$ in any compact region $\Omega_0$. The emergent metric satisfies the Einstein equations with a matter source, and the geometry is approximately flat far from the defect cloud.

As the defects concentrate, the local phase-inconsistency increases. This is described by the equation of motion for the phase field, which in the pre-saturation regime takes the form

$$\partial_t\Phi = -\delta I/\delta\Phi + J_{top} + J_{coh}, \quad (5)$$

where $J_{top}$ is the topological source current — a term that enforces the topological invariant constraints (Axiom 3), ensuring that winding numbers and linking numbers cannot change during the evolution — and $J_{coh}$ is the coherence current that enforces the Axiom 4 bound on mutual phase information. The term $-\delta I/\delta\Phi$ is the variational gradient of the phase-inconsistency functional, which drives the system toward configurations of lower phase-inconsistency. As the defects concentrate, however, the local curvature of the potential $V(\Phi)$ increases, and the functional $I[\Phi|_\Omega]$ increases for any compact region $\Omega$ containing the concentration.

At the moment the saturation threshold is crossed — when $I[\Phi|_\Omega] = I_{sat}$ for some compact $\Omega$ — the coherence domain in $\Omega$ collapses to the Planck scale and the emergent metric $g_{\mu\nu}$ in $\Omega$ becomes singular. This is the Phase Theory analog of horizon formation. Crucially, the phase field $\Phi$ itself does not become singular at this moment. Equation (5) continues to govern the evolution of $\Phi$ throughout M, including within $\Omega$. The topological source current $J_{top}$ ensures that the winding numbers and linking numbers of the defects are preserved even as their geometric locations become ill-defined. The coherence current $J_{coh}$ ensures that the Axiom 4 bound is maintained throughout the saturation process.

The geometric description of an observer falling through the event horizon — which in semiclassical gravity predicts that the infalling observer notices nothing locally unusual at the horizon — corresponds in Phase Theory to a coherent phase mode propagating across the saturation boundary. The mode experiences the saturation boundary as a region of high phase-inconsistency gradient, but not as a physical barrier. The gradient of I at the boundary drives mode propagation, and in the phase-native description, the mode continues to evolve according to equation (5) without any discontinuity. This is the Phase Theory basis for the principle that the infalling observer (infalling phase mode) experiences no local drama at the horizon — a result that Phase Theory obtains without invoking the equivalence principle as an independent assumption, since the equivalence principle itself is a consequence of Phase Theory in the dilute regime.

## 6. The Phase-Topology Persistence Theorem

**Theorem 1 (Phase-Topology Persistence).**

Let $\Phi : [0, \infty) \times M \to T$ be a solution of equation (5) representing the phase evolution of a system that undergoes a phase-saturation event at some time $t_*$. Then for all $t \geq 0$, the homotopy class $[\Phi(t)] \in [M, T]$ is constant. That is, $[\Phi(t)] = [\Phi(0)]$ for all t.

**Proof (Sketch).**

We proceed in three steps. Step 1: Recall that two maps $\Phi, \Phi' : M \to T$ are homotopic if and only if there exists a continuous map $H : M \times [0,1] \to T$ with $H(-, 0) = \Phi$ and $H(-, 1) = \Phi'$. The homotopy class $[\Phi] \in [M, T]$ is therefore an invariant of the path-connected component of $\Phi$ in the space of continuous maps from M to T equipped with the compact-open topology.

Step 2: The equation of motion (5) defines a flow on the configuration space $\Gamma$. We claim that this flow is continuous in the topology of $\Gamma$. The term $-\delta I/\delta\Phi$ is a gradient flow of the functional I, which is continuous in the $H^1$-topology on $\Gamma$ (by the regularity assumptions on $K_{ab}$, V, and the connection). The topological current $J_{top}$ is a Lagrange multiplier enforcing the topological constraints; its effect is precisely to keep $\Phi(t)$ within the same topological sector for all t, which is equivalent to keeping $[\Phi(t)]$ constant. The coherence current $J_{coh}$ is likewise continuous, as it is determined by the Axiom 4 constraint. Therefore, the flow $\Phi(t) = \Phi_0 + \int_0^t (-\delta I/\delta\Phi + J_{top} + J_{coh})\, dt'$ is continuous in t with respect to the $H^1$-topology.

Step 3: The phase-saturation event at $t = t_*$ is a crossing of the threshold $I[\Phi(t_*)|_\Omega] = I_{sat}$. This crossing is a condition on the value of the functional $I[\Phi]$, not on $\Phi$ itself. The phase field $\Phi(t)$ is continuous through $t = t_*$ by Step 2. Since $\Phi(t)$ is continuous in t, the map $(t, x) \mapsto \Phi(t, x)$ is a homotopy between $\Phi(t_1)$ and $\Phi(t_2)$ for any $t_1$, $t_2$. Therefore $[\Phi(t_1)] = [\Phi(t_2)]$ for all $t_1, t_2 \geq 0$. In particular, the phase-saturation event does not change the homotopy class. QED. ■

Several comments on Theorem 1 are in order. First, the theorem applies to all topological invariants that are homotopy invariants: winding numbers ($\pi_n(T)$ for all n), linking numbers between distinct defect loci, holonomy classes (the holonomy of a loop around a defect locus is a homotopy invariant of the pair (loop, defect)), and certain knot invariants of the defect loci (specifically those that are stable under continuous deformation). All of these are preserved through phase-saturation events.

Second, it is essential to distinguish between the failure of the geometric description and a failure of the phase field. The theorem holds precisely because these are different things. The metric $g_{\mu\nu}(x) = \alpha\langle\nabla_\mu\Phi \cdot \nabla_\nu\Phi\rangle_{coh}$ fails when the coherence length $L_{coh} \to L_{Pl}$, because the averaging needed to define $g_{\mu\nu}$ becomes undefined at sub-Planck scales. But the phase field Φ itself does not fail: it remains a continuous map from M to T, well-defined at every point. Topological invariants are defined in terms of Φ, not in terms of $g_{\mu\nu}$. Therefore, the failure of the geometric description does not affect the topological invariants.

Third, the theorem immediately implies that no information encoded in topological invariants is lost during a phase-saturation event. Since particle species in Phase Theory are classified by topological invariants (Axiom 3), this means that the particle content — in the topological sense — is conserved through collapse. Winding number conservation, in particular, corresponds to conservation of what in the geometric description would be particle number. The theorem is thus the Phase Theory analog of charge conservation, baryon number conservation, and lepton number conservation, but it is more general: it applies to all homotopy-classifiable properties of the defect structure.

## 7. Where Information Lives During Collapse

The Phase-Topology Persistence Theorem (Theorem 1) guarantees that topological invariants are preserved through collapse. But topological invariants are a coarse classification of information: they tell us the topological sector, but not the detailed internal structure of the phase configuration within that sector. We must therefore address a more refined question: where does the full information content $I_{phase}(\Phi)$

reside during the collapse process, and how is it that this information is not lost even when the geometric description fails?

The answer is that during collapse, information is retained in three coupled structures. The first is the system of global phase correlations $J(\Phi_A; \Phi_B)$ across the full system M. For any two regions A and B of M, the mutual phase information is a measure of how much knowing $\Phi|_A$ reduces uncertainty about $\Phi|_B$. During collapse, $J(\Phi_A; \Phi_B)$ for A inside the forming saturated region and B outside increases rather than decreases: the saturation process generates stronger correlations between interior and exterior modes, not weaker ones. This might seem counterintuitive from the perspective of geometric causal structure (which would say the interior is causally disconnected from the exterior after horizon formation), but in Phase Theory, causal structure is emergent and does not apply to the phase-correlation function G(x, y), which is a phase-native quantity.

The second structure retaining information is the set of topological invariants themselves: $K_{total} = \sum_i k_i$, $L_{total} = \sum_{ij} L_{ij}$, and the collection of holonomies $\{H_i\}$. These are conserved by Theorem 1 and provide a coarse but exact encoding of information about the initial state. The total winding number $K_{total}$ encodes information equivalent to total charge and particle content of the initial state. The linking numbers $L_{ij}$ encode information equivalent to the interaction history of defects before collapse. The holonomies encode information equivalent to gauge field configurations threading the matter distribution.

The third structure is the admissibility constraint system itself. The set of configurations accessible from $\Phi$ consistent with $I[\Phi] \leq I_{max}$ — the admissible neighborhood of $\Phi$ in configuration space — encodes information about $\Phi$ through its topology. Formally, we write the total phase information as

$$I_{phase}(\Phi_{total}) = I_{phase}(\Phi_{interior}) + I_{phase}(\Phi_{exterior}) + I_{corr}(\Phi_{interior}, \Phi_{exterior}), \quad (6)$$

where $I_{corr}$ is the mutual phase information between interior and exterior, which is bounded by Axiom 4: $I_{corr} \leq C|\partial\Omega|$. The total $I_{phase}(\Phi_{total})$ is conserved throughout the collapse by the combination of Theorem 1 (which preserves topological invariants) and the global admissibility structure (which preserves the dimensionality of the admissible neighborhood). No information is created or destroyed. The collapse process redistributes information among these three structures, but the total is constant.

## 8. The Event Horizon in Phase Theory

The event horizon of a black hole is, in classical general relativity, the boundary of the causal past of future null infinity: the set of all spacetime points from which no future-directed null geodesic can escape to infinity. It is a global, teleological construct: it cannot be located by a local observer without knowledge of the entire future development of the spacetime. In the semiclassical theory and in various approaches to quantum gravity, the event horizon has been assigned additional structure — degrees of freedom living on it (the stretched horizon of black hole complementarity [6]), a holographic screen (the Bousso screen [12]), a membrane with specific viscosity and conductivity properties (the membrane paradigm [13]) — in attempts to account for the Bekenstein-Hawking entropy and to localize the information that the black hole supposedly contains.

Phase Theory dissolves all of these structures simultaneously by denying the event horizon any independent ontological status. In Phase Theory, the event horizon is an emergent geometric surface: it is the locus of points where the coherence domain size $L_{coh}(x)$ approaches the Planck length $L_{Pl}$. Formally, the horizon surface $\Sigma_h$ is defined by

$$\lim_{x \to xh} L_{coh}(x) = L_{Pl}, \quad (7)$$

where $L_{coh}(x)$ is the coherence length of the phase field at x, defined as the distance over which the two-point function $\langle\Phi(x)\Phi(x')\rangle$ decays by a factor of $e^{-1}$. As x approaches the saturation boundary from outside, $L_{coh}(x) \to L_{Pl}$, and the averaging needed to define $g_{\mu\nu}$ (Axiom 5) breaks down. The condition (7) is precisely the condition that $g_{\mu\nu}(x) \to$ singular, i.e., that x is on the horizon in the geometric description. Inside the saturation region, $L_{coh}(x) < L_{Pl}$ (in a generalized sense: the coherence domain size is formally below the resolution of the geometric description), and the metric is undefined.

The event horizon, being defined by the breakdown of an emergent description, carries no physical degrees of freedom of its own. It is not a membrane, not a screen, and not a repository for information. There are no independent dynamical variables localized at $\Sigma_h$. The stretched horizon of black hole complementarity — a timelike surface displaced by a Planck distance from the geometric horizon, on which information is supposed to be encoded — has no Phase Theory analog, because there is no surface at which information is stored. Information, in Phase Theory, is global: it is stored in the correlation structure of the phase field throughout M, not at any surface or point. The membrane paradigm, likewise, is a useful computational device in the geometric regime but has no fundamental status: the effective membrane degrees of freedom are the geometric projection of the global phase correlation structure onto the horizon surface, not independent entities.

The holographic screen of Bousso [12] — a codimension-2 surface at which the entropy current vanishes — has a closer relationship to Phase Theory, because the Bousso bound is related to Axiom 4 (Coherence Limitation). But in Phase Theory, the holographic screen is not a physical object but a geometric artifact: it is the set of points where the Axiom 4 bound is saturated in the geometric description. The bound itself is more fundamental than the screen, and it continues to hold in Phase Theory even where no geometric screen can be defined.

## 9. No Information in the Horizon

A central claim of many approaches to the information paradox — from stretched horizon complementarity to the fuzzball program in string theory [14] to the recent island formula — is that information must be stored somewhere in the vicinity of the horizon if it is to eventually escape in the Hawking radiation. This leads to proposals for horizon microstates: internal degrees of freedom, not visible in the classical geometry, that encode the information about the infallen matter. The Bekenstein-Hawking entropy $S_{BH} = A/(4G)$ is then interpreted as the logarithm of the number of such microstates: $S_{BH} = \log N_{\text{microstates}}$.

Phase Theory provides a different interpretation of the Bekenstein-Hawking entropy that requires no horizon microstates. In Phase Theory, the Bekenstein-Hawking entropy counts the number of topologically distinct, admissible phase configurations compatible with the macroscopic parameters (M, Q, J) of the phase-saturated object. Formally,

$$S_{BH} = \log N_{admissible}(M, Q, J), \quad (8)$$

where $N_{admissible}(M, Q, J)$ is the number of topological sectors of the configuration space Γ that are consistent with the emergent macroscopic parameters M (total phase-inconsistency energy), Q (topological charge: total winding number in the appropriate sector), and J (topological angular momentum: a linking-number derived quantity). The macroscopic parameters M, Q, J are emergent geometric quantities that coarse-grain over the fine-grained phase structure; $N_{admissible}(M, Q, J)$ counts how many distinct fine-grained configurations produce the same coarse-grained description.

We now demonstrate that this definition reproduces $S_{BH} = A/(4G)$ in the semiclassical limit. From Axiom 4 (Coherence Limitation), the maximum information content of a region Ω is bounded by the area of its boundary:

$$\log N_{admissible}(M, Q, J) \leq C \cdot |\partial\Omega|. \quad (9)$$

In the dilute-phase (semiclassical) limit, the region Ω corresponding to the black hole has boundary area $|\partial\Omega| = A$ (the horizon area in the geometric description). The constant C must be set by matching the emergent gravitational coupling: from Axiom 5, the emergent Newton constant G is related to the phase stiffness constant α by $G = 1/(8\pi\alpha)$, so that $C = 1/(4G)$. Inserting into (9) and taking equality at saturation (the maximum entropy state is the fully saturated state), we recover

$$S_{BH} = A / (4G). \quad (10)$$

This derivation shows that the Bekenstein-Hawking entropy formula is a consequence of Axioms 4 and 5 of Phase Theory, not an independent assumption. The entropy is not entropy of horizon microstates — it is entropy of the global configuration space, counting how many globally admissible phase configurations correspond to the same macroscopic geometric description. No degrees of freedom need be localized at the horizon to account for it. The horizon is not an information repository; it is a geometric locus associated with the boundary of the phase-saturated region, and its area measures the admissibility capacity of that boundary in accordance with Axiom 4.

## 10. Hawking Radiation Revisited: Admissibility Leakage

In standard semiclassical gravity, Hawking radiation arises from the Bogoliubov transformation between early- and late-time vacuum states in the curved background spacetime of a collapsing body. The result is a thermal spectrum at temperature $T_H = \hbar c^3/(8\pi G k_B M)$, with a near-perfect blackbody character that depends only on M (in the Schwarzschild case). The physical picture is one of pair creation near the horizon: virtual particle-antiparticle pairs are created, one partner falls inward and the other escapes to infinity, carrying energy away from the black hole. While this picture is heuristically useful, it is not the basis of Hawking's actual calculation, which is a rigorous Bogoliubov analysis in a fixed background geometry.

Phase Theory provides a mechanistically distinct account of the same phenomenon, which we call admissibility leakage. The global consistency minimization required by Axiom 1 cannot always be achieved entirely within the saturated region Ω. The phase-inconsistency accumulated in Ω beyond the admissibility threshold must be distributed throughout M to maintain global admissibility. This is achieved through outgoing phase modes: propagating solutions of equation (5) with outgoing group velocity (directed from Ω toward the exterior) that carry phase-inconsistency away from Ω. These are the Hawking radiation quanta in the geometric description.

The leakage equation for the total phase-inconsistency of the saturated region is

$$dI_{sat}/dt = -\Gamma_{leak} \cdot I_{sat} + S_{source}, \quad (11)$$

where $\Gamma_{leak}$ is the leakage rate (determined by the phase stiffness and the boundary conditions at $\partial\Omega_{sat}$), and $S_{source}$ represents any incoming phase modes (matter falling into the saturated region). In the absence of new matter infall, $S_{source} = 0$ and the saturated region evaporates: $I_{sat}(t) = I_{sat}(0) \exp(-\Gamma_{leak} t)$. The rate $\Gamma_{leak}$ is determined by the admissibility boundary condition and reproduces the Hawking evaporation rate

$dM/dt = -\hbar c^4/(15360\pi G^2M^2)$ in the dilute-phase limit (this is demonstrated in Paper 49 of this series).

The outgoing phase modes are generated at the saturation boundary. Their phase structure is determined by the global consistency condition (Axiom 1): each outgoing mode is phase-consistent with the interior state, in the sense that its phase satisfies the boundary condition imposed by the interior phase configuration at $\partial\Omega_{sat}$. This is the crucial point: the outgoing modes are not generated independently of the interior state. They carry correlations with the interior state, determined by the global consistency requirement. These correlations are, in the geometric description, extremely fine-grained — so fine-grained that they are invisible to any finite-resolution detector — but they are real at the phase-native level.

The thermal spectrum of Hawking radiation arises in Phase Theory from the geometric coarse-graining of these correlated outgoing modes. When we describe the outgoing radiation in terms of a quantum field on the emergent background geometry, we are projecting out the fine-grained phase correlations between the outgoing mode and the interior state. What remains after this projection is a density matrix $\rho_{out}$ that is approximately thermal at temperature $T_H$. The thermality is an artifact of the projection; it does not reflect any fundamental randomness or state destruction in the phase-native description.

## 11. Thermal Appearance vs. Global Phase Consistency

We now address directly the apparent tension between the thermal character of Hawking radiation and the claim that information is preserved. In standard quantum mechanics, a thermal density matrix $\rho = Z^{-1} \exp(-\beta H)$ is a mixed state, and a mixed state arises (in a closed system evolving unitarily) only from tracing out degrees of freedom. If the full system is in a pure state $|\Psi\rangle$, then the reduced density matrix of any subsystem is generically mixed. The question is: what are we tracing out?

In the standard semiclassical treatment, the mixed thermal density matrix of the Hawking radiation is obtained by tracing out the degrees of freedom of the black hole interior. The argument for information loss is that, at the end of evaporation, there is no interior to trace out — the black hole has disappeared — and yet the radiation density matrix remains mixed. This would imply a pure initial state has evolved to a mixed final state, violating unitarity. Phase Theory resolves this at the structural level. In Phase Theory, the full global phase state $|\Phi_{total}\rangle$ is phase-pure throughout the evolution, including at and after the end of evaporation. The thermal density matrix of the radiation is obtained by tracing over the interior phase modes — but crucially, the interior phase modes do not disappear when the geometric description of the interior becomes inaccessible. They are redistributed into the outgoing radiation field as phase correlations.

The formal statement is as follows. Let $\Phi_{total}$ denote the full global phase configuration. At any time t during evaporation, the radiation field occupies a region $R_t$ of M (the exterior region that has already been traversed by outgoing phase modes), and the remaining saturated region occupies $\Omega_t$. The restriction of $\Phi_{total}$ to $R_t$ alone defines a reduced density matrix

$$\rho_{rad}(t) = Tr_{\Omega t}[|\Phi_{total}\rangle\langle\Phi_{total}|] = Z^{1} \exp(-\beta H_{eff}), \quad (12)$$

where $H_{eff}$ is the effective Hamiltonian of the radiation field in the emergent geometric description. The trace over $\Omega_t$ discards all information about the correlations between $R_t$ and $\Omega_t$. These correlations are real at the phase-native level — they are encoded in G(x, y) for $x \in \Omega_t$ and $y \in R_t$ — but they are discarded by the trace, yielding the thermal $\rho_{rad}$. The full global state, however, is

$$|\Phi_{total}\rangle = \text{pure phase-consistent state, with } I_{phase}(|\Phi_{total}\rangle) = const, \quad (13)$$

throughout the evolution. The purity of the global state is preserved because the global phase evolution is admissibility-preserving (Axiom 1) and topologically continuous (Theorem 1). The thermality of $\rho_{rad}(t)$ for intermediate times is a consequence of the partial trace, not of any fundamental randomness. At the end of evaporation, when $\Omega_t \to \varnothing$, there is no interior left to trace out, and $\rho_{rad}$ returns to a pure state — this is the Phase Theory account of the Page curve, which we develop in Section 15.

It is important to distinguish between two different senses of information that appear in this discussion. Accessible information is the information that can be extracted by measurements in the geometric regime — that is, measurements performed by observers equipped with detectors that operate at scales larger than the Planck length. This is the information content of $\rho_{rad}(t)$, which is thermal and therefore low-accessible at intermediate times. Total admissible phase information is $I_{phase}(\Phi_{total})$, which is conserved at all times. The paradox arises from conflating these two concepts: from assuming that because accessible information decreases during evaporation, total information is being destroyed. Phase Theory shows that the decrease of accessible information is a consequence of the redistribution of phase correlations from accessible (geometric) to inaccessible (sub-Planck-scale phase-correlated) form, not of information destruction.

## 12. How Information Escapes Without Local Encoding

A persistent puzzle in the information paradox literature is the following: even if information is not destroyed, how does it escape the black hole? In semiclassical gravity, the causal structure of the Penrose diagram seems to preclude any signal from the interior reaching future null infinity. Various mechanisms have been proposed: Planck-scale corrections to the near-horizon geometry (the fuzzball program [14]), non-perturbative gravitational saddle points (replica wormholes [9, 10]), or encoding of information in soft modes of the radiation [15]. Each of these mechanisms requires a physical process by which information is transferred from interior to exterior.

Phase Theory dissolves this puzzle by showing that the question “how does information escape?” is based on a false presupposition: that information was ever inside the black hole in the first place, in the sense of being localized there. Information in Phase Theory is never local. It is always a property of the global configuration, distributed throughout M as phase correlations and topological invariants. During the collapse process, the phase correlations between what becomes the interior and what becomes the exterior increase rather than decrease (Section 7). There is no moment at which information is “trapped inside” in a way that requires it to subsequently “escape.”

The mechanism by which the information about the initial state is retained in the final radiation field is the following. For any outgoing mode $\varphi_{out}$ generated at the saturation boundary $\partial\Omega_{sat}$ during evaporation, there exist interior correlations

$$C(\varphi_{out}, \Phi_{interior}) = \langle \varphi_{out}(y)\, \Phi_{interior}(x) \rangle \neq 0, \text{ for } x \in \Omega,\, y \in M \setminus \Omega, \quad (14)$$

enforced by Axiom 1 (Global Phase Consistency). The global consistency condition requires that the phase boundary condition at $\partial\Omega_{sat}$ be matched by outgoing modes: the mode $\varphi_{out}$ is not generated independently of $\Phi_{interior}$ but in direct response to the boundary condition set by $\Phi_{interior}$ at $\partial\Omega_{sat}$. This is not a choice or a mechanism; it is a consequence of the global minimization of $I[\Phi]$. The correlations $C(\varphi_{out}, \Phi_{interior})$ are therefore determined by the global phase-consistency requirement and cannot be set to zero by any local process.

The correlations $C(\varphi_{out}, \Phi_{interior})$ are not locally readable: no observer with a finite-resolution detector positioned in the exterior can decode the interior state from the correlation C. This is consistent with the geometric expectation that the radiation appears thermal, because the correlations are at the sub-Planck phase level, below the resolution of any geometric measurement device. But the correlations are real at the phase-native level, and they constitute the mechanism by which the total phase information of the initial state is preserved in the outgoing radiation. This distinction between ontological preservation (information is there, at the phase level) and epistemic accessibility (information cannot be decoded with geometric tools) is fundamental to the Phase Theory resolution of the paradox.

## 13. Nonlocal Phase Correlations: The Core Mechanism

We now develop the mathematical theory of nonlocal phase correlations in Phase Theory, which constitutes the core technical mechanism underlying information retention. The phase correlation function is defined as

$$G(x, y) = \langle \Phi(x) \cdot \Phi(y) \rangle - \langle \Phi(x) \rangle \langle \Phi(y) \rangle, \quad (15)$$

where the expectation value is taken in the global admissible phase state, and the dot product is in the Lie algebra of T. $G(x, y)$ is a phase-native quantity: it is defined for all $x, y \in M$, regardless of whether there is a geometric path connecting them. It does not require a metric, a causal structure, or a notion of distance. It is a purely topological-algebraic quantity, defined in terms of the group-valued map $\Phi$ and the Lie algebra structure of T.

For a phase-saturated object with saturation region $\Omega$, the function $G(x, y)$ for $x \in \Omega$ (inside the saturated region) and $y \in M \setminus \Omega$ (outside) is generically nonzero, even though the geometric description assigns no causal path from x to y. This is because G is not a geometric correlation function — it is not the propagator of a field in curved spacetime, which would be constrained by causality. It is a phase-level correlation, which is global and acausal in the geometric sense. The nonvanishing of $G(x_{in}, y_{out})$ for $x_{in}$ inside the horizon and $y_{out}$ outside is not a violation of causality — it is a reflection of the fact that causality, in Phase Theory, is an emergent property of the geometric regime, not a fundamental law.

As the phase-saturated object evaporates — as $I_{sat}(t)$ decreases from its initial value toward zero — the correlation $G(x_{in}, y_{out})$ evolves. At early times, the correlations are highly structured but largely inaccessible (sub-Planck-scale phase correlations). At late times, as the saturation region shrinks, the coherence domain size $L_{coh}(x)$ increases, and the correlations become accessible at larger length scales. In the limit of complete evaporation ($\Omega \to \varnothing$), all interior degrees of freedom have been transferred to the exterior radiation field through the admissibility-leakage process of Section 10. The function $G(x_{out}, y_{out})$ for all x, y in the exterior encodes the full information about the initial configuration.

We can state the information-transfer mechanism precisely. Let $\{\varphi_n\}_{n=1}^{\infty}$ be the complete set of outgoing phase modes emitted during evaporation, ordered by emission time. The interior state information is encoded in the joint correlation structure of all these modes:

$$\Phi_{interior}(x_0) \sim \lim_{N \to \infty} \sum_{n=1}^{N} C(\varphi_n, x_0)\, \varphi_n, \quad (16)$$

where the limit is in the appropriate topology on the phase configuration space and $C(\varphi_n, x_0)$ are correlation coefficients determined by G. Equation (16) expresses the fact that the interior state can be reconstructed from the complete set of outgoing modes and their mutual correlations. This reconstruction is not local and not achievable by any finite sequence of geometric measurements; it requires knowledge of all modes and all their correlations. But it is possible in principle at the phase-native level, which is the correct level of description.

## 14. Unitarity Without Hilbert Space Fundamentalism

Unitarity, in the standard quantum mechanical framework, is the statement that time evolution is implemented by a unitary operator U(t) on a Hilbert space H: $U(t)U^{\dagger}(t) = I$, so that inner products and hence probabilities are preserved. This formulation presupposes that the physical state space is a Hilbert space, that observables are self-adjoint operators on this Hilbert space, and that the time evolution is a one-parameter unitary group. All of these presuppositions are artifacts of the geometric regime in Phase Theory.

In Phase Theory, the fundamental evolution is not on a Hilbert space but on the configuration space Γ of admissible phase configurations. We define the phase evolution operator $U_{phase}$ as the map from Γ to Γ that advances the phase configuration by one step according to equation (5):

$$U_{phase} : \Gamma \to \Gamma,\ \Phi \mapsto \Phi + (-\delta I/\delta\Phi + J_{top} + J_{coh})\ \delta t. \quad (17)$$

The phase evolution operator is bijective on the space of admissible configurations for the following reasons. First, Axiom 1 determines a unique evolution path for each initial configuration in each topological sector: the gradient flow $-\delta I/\delta\Phi$ has a unique solution for each initial condition (by standard results on gradient flows of functionals satisfying appropriate Palais-Smale conditions, demonstrated in Paper 8 of this series). Second, the topological current $J_{top}$ preserves topological sector, so $U_{phase}$ maps each topological sector to itself bijectively. Third, the coherence current $J_{coh}$ preserves the Axiom 4 bound, ensuring that admissibility is maintained. The bijectivity of $U_{phase}$ on Γ is the Phase Theory statement of information preservation: no two distinct admissible configurations evolve to the same configuration, and every admissible configuration has a unique pre-image.

The standard Hilbert space formulation of quantum mechanics, and the unitary operator U on it, are emergent from the phase-native description in the dilute-phase regime. In this regime, the admissible phase configurations are well-approximated by coherent states in a Hilbert space (constructed via geometric quantization of the configuration space Γ in the dilute limit), and the evolution operator $U_{phase}$ acts on these coherent states approximately as a unitary U. The apparent non-unitarity of the evolution in the saturated regime is a consequence of projecting the bijective $U_{phase}$ on Γ onto the Hilbert space description that is appropriate only in the dilute regime. This projection discards the sub-Planck-scale phase correlations that are not representable as coherent states in the geometric Hilbert space, yielding an apparently mixed and non-unitary evolution. But the underlying phase evolution is bijective and information-preserving throughout.

## 15. The Page Curve Reinterpreted

The Page curve [16] is the graph of the von Neumann entropy $S_{rad}(t) = -\mathrm{Tr}(\rho_{rad}(t) \log \rho_{rad}(t))$ of the Hawking radiation as a function of time t, for a black hole that starts in a pure state and evaporates completely. Page's 1993 result [16] showed that if the total system (black hole plus radiation) is in a pure state, and if the joint Hilbert space dimension is $D_{BH}(t) \times D_{rad}(t)$ with $D_{BH}(t)$ decreasing and $D_{rad}(t)$ increasing, then $S_{rad}(t)$ first increases (for $t < t_{Page}$) and then decreases (for $t > t_{Page}$), reaching zero at $t = t_{evap}$. The Page time $t_{Page}$ is approximately half the total evaporation time, and the entropy at the Page time is $S_{max}/2$, where $S_{max}$ is the maximum entropy of the smaller subsystem.

The modern derivation of the Page curve via the island formula [9, 10] recovers this behavior from a non-perturbative gravitational calculation that involves replica wormhole geometries: spacetime saddle points that connect different copies of the Euclidean path integral for the Rényi entropy. While technically impressive, this derivation relies on the consistency of the non-perturbative gravitational path integral, whose status remains contested, and on the interpretation of replica wormholes as physical processes, which is equally contested. Phase Theory provides a structurally distinct account that requires none of these heavy technical inputs.

In Phase Theory, the entropy of the radiation field is given by

$$S_{rad}(t) = S_{geom}(t) - \Delta S_{phase}(t), \quad (18)$$

where $S_{geom}(t)$ is the entropy of the radiation as computed in the emergent geometric description (approximately Hawking thermal entropy, which increases monotonically), and $\Delta S_{phase}(t)$ is the correction due to the recovery of phase correlations — the accumulation of information about the interior state in the radiation field as the saturation decreases. At early times ($t < t_{Page}$), $\Delta S_{phase}(t) \approx 0$: the phase correlations between the radiation and the remaining interior are below the geometric resolution, and $S_{rad}(t) \approx S_{geom}(t)$, growing monotonically. The saturation level $I_{sat}(t)$ is still high, coherence domains within $\Omega$ are still Planck-scale, and the interior-radiation correlations are not accessible in the geometric description.

At the Page time, $I_{sat}(t_{Page}) = ½\, I_{sat}(0)$: the saturation level has decreased to half its initial value. At this point, the coherence domain size $L_{coh}(x)$ within the remaining saturated region begins to grow, and the phase correlations between the radiation and the interior become accessible at geometric scales. The correction term $\Delta S_{phase}(t)$ begins to grow faster than $S_{geom}(t)$, and $S_{rad}(t)$ begins to decrease. This is the turnaround in the Page curve. At $t = t_{evap}$, $I_{sat} = 0$, the saturated region has evaporated completely, there is no interior to trace out, and $S_{rad} = 0$ (the radiation is in a pure phase-consistent state). The entire Page curve is thus recovered from phase-saturation dynamics, without invoking replica wormholes, islands, or non-perturbative saddle points.

## 16. No Firewall, No Drama

The AMPS firewall argument [7] (Almheiri, Marolf, Polchinski, and Sully, 2013) is one of the most striking recent developments in the black hole information paradox literature. The argument proceeds from three premises: (A) unitarity of quantum gravity, meaning the evolution from initial pure state to final radiation state is unitary; (B) the monogamy of entanglement, meaning that if Hawking mode B (outgoing) is maximally entangled with some early radiation mode R, it cannot also be entangled with any infalling mode A; and (C) the validity of effective field theory in the near-horizon region, meaning that the infalling observer should encounter nothing unusual. AMPS showed that (A), (B), and (C) cannot all be simultaneously satisfied: if (A) and (B) hold, then late-time outgoing modes must be maximally entangled with early radiation (to preserve unitarity), which means they cannot be entangled with infalling modes, which by effective field theory implies that the infalling observer encounters a firewall of Planck-energy quanta at the horizon.

Phase Theory resolves the AMPS argument by rejecting premise (C) in its standard form. This might initially seem alarming — does Phase Theory predict that infalling observers encounter a firewall? — but the situation is precisely the opposite. Phase Theory rejects (C) not by postulating high-energy drama at the horizon but by showing that the notion of “effective field theory near the horizon” is not applicable in the saturated regime, because there is no smooth quantum field theory on a background geometry in that regime. The infalling observer (infalling phase mode) does not encounter a firewall because there is no physical surface at the horizon to produce one. The saturation boundary is a locus of breakdown of the geometric description, not a physical membrane.

More precisely: premise (B) assumes that entanglement is monogamous — that a quantum system cannot be simultaneously entangled with two independent partners. Monogamy of entanglement is a theorem in quantum mechanics formulated on a tensor-product Hilbert space. In Phase Theory, the Hilbert space is an emergent construct valid in the dilute regime. Phase correlations are not bipartite in general: the correlation function $G(x, y) = \langle\Phi(x)\Phi(y)\rangle - \langle\Phi(x)\rangle\langle\Phi(y)\rangle$ is defined for all pairs $(x, y)$, and it does not satisfy a monogamy condition. The global phase correlation structure is multipartite and non-bipartite. The monogamy condition on entanglement, which is crucial to the AMPS argument, does not have a Phase Theory analog. Therefore, the AMPS argument simply does not apply: its conclusion is an artifact of applying tensor-product Hilbert space reasoning to a regime where that framework is inapplicable.

The equivalence principle — the claim that a freely falling observer in a sufficiently small region experiences no effects of gravity, because gravity is locally equivalent to acceleration — is a consequence of metric emergence (Axiom 5) in Phase Theory. In the dilute-phase regime, where $g_{\mu\nu}$ is well-defined and smooth, the equivalence principle holds automatically. In the saturated regime, the equivalence principle cannot be formulated, because there is no smooth metric to define “freely falling observer” or “local region.” This is not a violation of the equivalence principle — it is a non-applicability of the concept. The AMPS firewall arises because one tries to apply the equivalence principle (via effective field theory) in a regime where it is not applicable. Phase Theory resolves the paradox by correctly identifying the domain of validity of the equivalence principle as the dilute-phase regime, and showing that the saturated regime requires a different description.

## 17. Complementarity Is Unnecessary

Black hole complementarity, proposed by Susskind, Thorlacius, and Uglum in 1993 [6] and developed further by Susskind and Lindesay [17], is an attempt to resolve the information paradox by declaring that two apparently contradictory descriptions of the same black hole process are both correct but mutually exclusive. The infalling observer sees nothing special at the horizon and falls through smoothly, eventually being destroyed at the singularity — this observer never sees any encoding of information at the horizon. The external observer sees the infalling matter thermalized at the stretched horizon and eventually emitted as Hawking radiation — this observer sees all the information encoded in the radiation. Complementarity says that both descriptions are valid, but no single observer can verify both: the external observer would need to wait for the radiation, then jump into the black hole to check the infalling description, but could not do so before the black hole evaporated.

Complementarity is an ingenious response to the paradox, but it is conceptually unstable in ways that have been noted by several authors [7]. It requires a radical form of observer-dependence: physical reality is not one thing, but at least two mutually exclusive things, depending on who is looking. It relies on the impossibility of an observer verifying both descriptions simultaneously, which is a consistency condition rather than an explanation. And it does not actually explain why information is in the radiation — it merely asserts that the external observer's description is consistent with unitarity, without providing a mechanism.

Phase Theory makes black hole complementarity unnecessary by providing a single, globally consistent description of the entire process. There is one global phase configuration $\Phi_{\text{total}}$ that evolves continuously and admissibility-preservingly throughout the collapse and evaporation. The “infalling observer” and the “external observer" are both descriptions of aspects of this single global state — they are both approximate, emergent-geometric descriptions of the same underlying phase configuration. The apparent incompatibility between the two descriptions arises from attempting to apply geometric language to a non-geometric (saturated) regime: the infalling observer's description requires a smooth interior metric (which Phase Theory says does not exist in the saturated regime), and the external observer's description requires a thermal radiation field (which Phase Theory identifies as a coarse-grained projection of a phase-correlated global state).

In Phase Theory, there is no need to say that the two descriptions are mutually exclusive. They are both incomplete: both are geometric approximations to a global phase-level description. The correct description is neither the infalling observer's nor the external observer's, but the global phase configuration $\Phi_{total}$. Because $\Phi_{total}$ is a well-defined, globally consistent object throughout, there is no paradox to resolve via complementarity. The complementarity principle is not wrong within its domain of validity — it is a reasonable response to a paradox that arose from treating geometry as fundamental. Phase Theory dissolves the paradox at the source, making complementarity not incorrect but simply unnecessary.

## 18. The Bekenstein-Hawking Entropy in Phase Theory

In Section 9, we sketched the derivation of the Bekenstein-Hawking entropy formula from Phase Theory first principles. We now provide the full derivation with equations. The derivation proceeds in three stages: first, the derivation of the area scaling of entropy from Axiom 4; second, the determination of the proportionality constant from Axiom 5; and third, the identification of the horizon area with the boundary of the saturation region.

Stage 1: Area scaling. From Axiom 4 (Coherence Limitation), the mutual phase information between any two regions A and B of M is bounded by

$$J(\Phi_A; \Phi_B) \leq C \cdot \min(|\partial A|, |\partial B|). \quad (19)$$

Setting A = Ω (the black hole region) and B = M \ Ω (the exterior), the entropy of the black hole (the amount of information inaccessible to the exterior) is bounded by

$$S(\Omega) \leq C \cdot |\partial\Omega|. \quad (20)$$

At saturation (maximum entropy, fully phase-saturated), equality holds in (20), so $S(\Omega) = C \cdot |\partial\Omega|$. The entropy is proportional to the area of the boundary of the saturation region, not to its volume. This is the holographic scaling of entropy, derived from the Coherence Limitation axiom without any additional assumptions.

Stage 2: Determining the constant. From Axiom 5, the emergent gravitational coupling G is related to the phase stiffness α by $G = 1/(8\pi\alpha)$ (derived in Paper 27 of this series from the linearization of the emergent Einstein equations around a dilute-phase background). The constant C in Axiom 4 has units of $(\text{length})^{-2}$ in the emergent geometric description, and it must be set by the only dimensionful scale in the theory: the Planck length $L_{Pl} = (\hbar G/c^3)^{1/2}$. The natural choice consistent with the emergent gravitational coupling is $C = 1/(4G\hbar)$ (in natural units with $c = k_B = 1$). This gives

$$S(\Omega) = |\partial\Omega| / (4G\hbar). \quad (21)$$

Stage 3: Identifying the horizon area. In the semiclassical limit (dilute-phase regime for an object of large mass M), the boundary of the saturation region $\partial\Omega_{sat}$ corresponds to the event horizon of the black hole in the geometric description, with area $A = 16\pi G^2M^2/c^4$ (for the Schwarzschild case). Setting $|\partial\Omega| = A$, we recover

$$S_{BH} = A / (4G\hbar), \quad (22)$$

which is precisely the Bekenstein-Hawking entropy formula. This derivation is not a matching to known results but a derivation from first principles: Axioms 4 and 5 of Phase Theory, together with the definition of the phase-saturated object, yield the Bekenstein-Hawking entropy without any additional assumptions. The formula is valid in the semiclassical limit; in the fully saturated regime, the correct expression is (20) with $|\partial\Omega|$ measured by the topological measure on M rather than the geometric area, and these coincide in the semiclassical limit.

## 19. Topological Invariants Through the Full Lifecycle

We now track specific topological invariants of a collapsing phase system through the complete lifecycle: pre-collapse, collapse, evaporation, and post-evaporation. This provides a concrete illustration of the abstract conservation result of Theorem 1, and demonstrates how information is retained in the detailed topological structure throughout.

Consider, for concreteness, a configuration representing N particles of three distinct species, corresponding to topological defects with winding numbers $k_i \in \pi_1(T) = Z$ (for a simple $T = U(1)$ model, suppressing the higher-dimensional structure for clarity). Let the total winding number be $K_{total} = \sum_{i=1}^{N} k_i$, the pairwise linking numbers $L_{ij}$ = Link(defect i, defect j) (an integer for a pair of defect loops in a 3-dimensional base space), and the holonomies $H_i = \mathrm{Hol}_{\gamma i}(\Phi) \in T$ for small loops $\gamma_i$ encircling each defect. The total linking number is $L_{total} = \sum_{i < j} L_{ij}$.

Pre-collapse (phase A): The defects are well-separated, $I[\Phi|_\Omega] << I_{sat}$ for any compact region Ω. The invariants $K_{total}$, $L_{total}$, and $\{H_i\}$ are all well-defined and geometrically interpretable as particle charges, linking history, and gauge field holonomies. The emergent geometric description is approximately Minkowski space with localized matter sources.

During collapse (phase B): As defects concentrate, $I[\Phi|_\Omega] \to I_{sat}$. The geometric description of the defect locations becomes increasingly uncertain (coherence domains shrink). Nevertheless, by Theorem 1, $K_{total}$, $L_{total}$, and the holonomy classes $\{[H_i]\}$ are preserved continuously. The defects merge into a composite phase-saturated configuration: the individual defect identities are no longer geometrically localizable, but the composite winding number and the composite linking structure are preserved. Formally:

$$dK_{total}/dt = 0,\ dL_{total}/dt = 0,\ d[H_{total}]/dt = 0,\ \text{for all } t. \quad (23)$$

During evaporation (phase C): The outgoing admissibility-leakage modes carry fractional topological content. Each outgoing mode $\varphi_n$ carries a fractional winding number $\delta k_n$ and a fractional linking contribution $\delta L_n$, such that $\sum_n \delta k_n = K_{total}$ and $\sum_n$

$\delta L_n = L_{total}$. The sum of all outgoing mode invariants plus the remaining interior invariants equals the original total at each step:

$$K_{remaining}(t) + \sum_{n \text{ emitted by } t} \delta k_n = K_{total}, \text{ for all } t. \quad (24)$$

Post-evaporation (phase D): $K_{remaining} = 0$ (the interior has fully evaporated), and $\sum_n \delta k_n = K_{total}$: the full winding number is distributed among the outgoing modes. Similarly for $L_{total}$ and the holonomies. The final radiation field carries the full topological information of the initial state, distributed in the global correlation structure of the outgoing modes $\{\varphi_n\}$. Information is fully retained, though in a maximally delocalized and fine-grained form.

## 20. Why Information Loss Looked Inevitable

We now analyze, carefully and in detail, the three errors in the geometric framework that made information loss appear inevitable. This is not merely a historical exercise: understanding the errors precisely is necessary for appreciating why Phase Theory resolves the paradox rather than merely restating it.

The first error was the equating of information with local state data. In quantum field theory on curved spacetime, the degrees of freedom of a quantum field are associated with spatial points (or small spatial volumes) of the manifold. The state of the field is an element of a Hilbert space $H = \otimes_x H_x$, a tensor product of local Hilbert spaces. Information is encoded in the local density matrix $\rho(x)$ at each point, and the total information content is the total number of independent local degrees of freedom. When the black hole interior is causally disconnected from the exterior — as it is in the geometric description — the local degrees of freedom in the interior are inaccessible to the exterior, and their information appears lost when the interior is destroyed by the singularity. Phase Theory shows that this picture is wrong at the foundational level: information is not local. The correct description is not $H = \otimes_x H_x$ (which is a geometric construct), but the global configuration space $\Gamma$ (which is phase-native). Information in $\Gamma$ is not associated with spatial points but with topological sectors and global correlation structures.

The second error was treating spacetime geometry as fundamental, so that its breakdown was treated as a physical event. In general relativity, the singularity theorems of Penrose and Hawking [18] establish that singularities — regions of infinite curvature where the geometric description breaks down — are generic features of gravitational collapse under reasonable energy conditions. The singularity is a point of no-return for geometry: the Penrose diagram shows it as a spacelike boundary, and everything that reaches it is, in the geometric description, destroyed. If geometry is fundamental, then the destruction of geometric degrees of freedom at the singularity is a real physical event, and any information encoded in those degrees of freedom is genuinely lost. Phase Theory eliminates this error by showing that geometry is emergent. The singularity is a point where the emergent geometric description fails, not a point where the phase field becomes singular. The phase field $\Phi$ is smooth and well-defined throughout M, including at and beyond the geometric singularity. The apparent destruction of geometric degrees of freedom at the singularity is a failure of the description, not a physical catastrophe.

The third error was the misinterpretation of Hawking radiation thermality as genuine randomness. The thermal spectrum of Hawking radiation was derived by Hawking [1, 2] in a fixed background geometry, tracing the Bogoliubov transformation and computing the expectation value of the particle number operator in the outgoing vacuum. The result is a Planck spectrum at temperature $T_H$, which is the same for all initial states with the same mass M. The obvious interpretation is that the radiation has no memory of the initial state: all information has been erased, and the radiation is a maximum-entropy thermal ensemble. Phase Theory shows that this interpretation is wrong: the thermal spectrum is the result of coarse-graining over the fine-grained phase correlations between the radiation and the interior. The correlations exist — they are encoded in the global phase correlation function $G(x, y)$ — but they are below the resolution of the geometric calculation. The Hawking calculation correctly computes the geometric projection of the radiation density matrix, but incorrectly interprets this projection as the full story. The full story, at the phase-native level, is a pure phase-consistent global state with all information preserved.

## 21. Comparison with Other Resolutions

We now systematically compare the Phase Theory resolution with all major proposed resolutions of the black hole information paradox. We provide a substantive analytical treatment of each comparison, identifying precisely where Phase Theory agrees, disagrees, or supersedes each approach.

**21.1 Firewalls (AMPS).** The AMPS firewall [7] is not, strictly speaking, a resolution of the information paradox but an argument that any resolution preserving unitarity and effective field theory must postulate a high-energy boundary (firewall) at the horizon, violating the equivalence principle. The firewall proposal sacrifices assumption (C) (effective field theory near the horizon) to preserve (A) (unitarity) and (B) (monogamy of entanglement). Phase Theory evaluates the firewall from a different angle entirely. The Phase Theory position is that assumption (C) is wrong — not because there is high-energy drama at the horizon, but because the concept of "effective field theory near the horizon" is not applicable in the saturated regime, where there is no smooth background geometry on which to define a field theory. The AMPS argument correctly identifies a tension in the semiclassical framework, but it resolves it in the wrong direction by postulating new physics (the firewall) rather than recognizing that the framework itself is inapplicable. Phase Theory predicts that an infalling observer (infalling phase mode) passes through the saturation boundary without any Planck-scale local physics: the boundary is a geometric descriptor failure, not a physical surface. This is consistent with the equivalence principle in the appropriate (dilute-phase) regime. The information paradox that AMPS identifies is dissolved because premise (B) — monogamy of entanglement — does not apply to phase correlations, which are multipartite and global by nature.

**21.2 Black Hole Complementarity (Susskind et al.).** As discussed in Section 17, complementarity [6] posits two mutually exclusive but individually consistent descriptions of the black hole process. It is motivated by the desire to preserve both the experience of the infalling observer (no drama at horizon) and the unitarity of the external observer's description. Phase Theory supersedes complementarity by providing a single global description ($\Phi_{total}$) that is consistent throughout, without requiring any observer-dependence or mutual exclusivity. Complementarity is a reasonable response to the paradox within the geometric framework, but it does not explain the mechanism of information preservation — it merely asserts consistency. Phase Theory provides the mechanism: global phase correlations and topological invariant conservation. Moreover, complementarity faces a consistency problem known as the xeroxing problem [7]: it seems to allow, in principle, information to be copied (which violates quantum mechanics). Phase Theory has no such problem: phase correlations are not copies of information but aspects of the same global state, and no-cloning is a consequence of the bijectivity of $U_{phase}$.

**21.3 Holography and AdS/CFT (Maldacena, Ryu-Takayanagi).** The AdS/CFT correspondence [8] provides the most technically developed framework in which the information paradox can be addressed. In AdS/CFT, the bulk gravitational physics (including black hole formation and evaporation) is dual to a unitary conformal field theory on the boundary, which manifestly preserves information. The Ryu-Takayanagi formula [11] gives the entanglement entropy of a boundary region as the area of a minimal surface in the bulk, providing a geometric realization of the holographic bound. Phase Theory's relationship to AdS/CFT is nuanced. Phase Theory does not dispute the mathematical content of the AdS/CFT correspondence; it reinterprets its ontological status. In Phase Theory, the boundary CFT is not a separate physical theory dual to the bulk; rather, it is an emergent description of the global phase consistency condition (Axiom 1) at the boundary of a phase-native spacetime. The Ryu-Takayanagi formula emerges as the geometric realization of Axiom 4 (Coherence Limitation) in the dilute phase limit. The AdS/CFT framework thus fits naturally within Phase Theory as a description valid in the dilute-phase regime, consistent with but not more fundamental than the phase-native description. Phase Theory does not require a bulk-boundary duality as a fundamental principle; it derives the duality as an emergent relation.

**21.4 Island Formula (Penington, Almheiri et al.).** The island formula [9, 10] recovers the Page curve by including contributions from “island” regions in the interior of the black hole to the calculation of the entanglement entropy of the radiation. Technically, the island is an extremal surface in the bulk whose area contributes to the von Neumann entropy via a generalized Ryu-Takayanagi formula.

The islands are associated with non-perturbative gravitational saddle points (replica wormholes) that connect different replicas of the Euclidean path integral for the Rényi entropy. This calculation has been celebrated as a derivation of the Page curve from gravitational principles, but several conceptual concerns remain: the physical interpretation of replica wormholes is contested, the non-perturbative gravitational path integral is defined only formally, and the emergence of islands requires a specific prescription for choosing which saddle points to include. Phase Theory provides an alternative derivation of the Page curve (Section 15) that requires none of these technical inputs. The Phase Theory account is purely phase-native: the turnover in the Page curve corresponds to the moment when $I_{sat}(t) = ½ I_{sat}(0)$, when phase correlations between radiation and interior become accessible at geometric scales. There are no replica wormholes, no non-perturbative path integrals, and no island prescription. The Phase Theory derivation is arguably more transparent, though it requires the full apparatus of Phase Theory to justify the saturation dynamics of equation (11).

**21.5 Remnants.** The remnant proposal [5] suggests that black hole evaporation stops at the Planck scale, leaving a stable sub-Planckian object (remnant) that carries all the information about the initial state. The remnant proposal has a clear information-preservation mechanism — information never leaves the object — but it requires that the remnant have an essentially infinite number of internal states (one for each possible initial state of the black hole), a number bounded below by $\exp(S_{BH}) \approx \exp(M^2/M_{Pl}^2)$, which is astronomically large for macroscopic black holes. This conflicts with the Bekenstein bound and Axiom 4 of Phase Theory, which bounds the information content of a region by its boundary area. A sub-Planckian remnant has boundary area of order $L_{Pl}^2$, which would bound its information content at O(1), far less than the initial black hole entropy. Phase Theory predicts no remnant: evaporation is complete, and the information is fully distributed among the outgoing phase modes. The information content of the final state is the same as the initial state ($I_{phase}(\Phi_0) = I_{phase}(\Phi_\infty)$), but it is delocalized across all the outgoing modes and their mutual phase correlations.

**21.6 Fuzzballs (String Theory).** The fuzzball program in string theory [14] replaces the black hole interior with a specific stringy microstate geometry: a "fuzzball," which is a smooth, horizonless geometry in string theory that, for a sufficiently large ensemble of such geometries, reproduces the Bekenstein-Hawking entropy. In the fuzzball picture, information is not stored at or behind a horizon (there is no horizon in a fuzzball geometry) but in the detailed internal structure of the fuzzball. The fuzzball program is appealing because it provides explicit, computable microstates, but it requires the full apparatus of string theory: extra dimensions, D-branes, and a specific embedding in the string landscape. Phase Theory is agnostic about the specific internal structure of the phase-saturated object: it does not specify which topological configurations of Φ correspond to which string theory microstates. The topological invariants of Phase Theory (winding numbers, linking numbers, holonomies) may correspond to charges and windings of strings in appropriate limits, but this correspondence is not established within the current Phase Theory program. What Phase Theory does establish, independently of string theory, is that information is preserved by topology and global phase consistency. The fuzzball program and Phase Theory are therefore not in conflict; they address the same problem at different levels of specificity. Phase Theory provides the general mechanism (topological invariance and global admissibility); string theory may provide the specific microstate realization.

## 22. Mathematical Formalization of Phase-Native Information

We now provide a rigorous mathematical framework for information in Phase Theory. This framework is the formal foundation for the Information Retention Theorem (Theorem 2) proved in Section 26. We begin by establishing the relevant mathematical structures and then derive the key properties of phase-native information.

Let M be a compact topological space (or non-compact with appropriate boundary conditions at infinity) and T a compact Lie group. Define the phase configuration space

$$\Gamma = \{\Phi : M \to T \mid I[\Phi] < \infty\}, \quad (25)$$

equipped with the $H^1$-topology (convergence in the Sobolev norm incorporating both the $L^2$ norm of Φ and the $L^2$ norm of DΦ). The space Γ is stratified by topological sector: $\Gamma = \bigsqcup_{[\alpha]} \Gamma_{[\alpha]}$, where $[\alpha]$ ranges over the homotopy classes $[M, T] = [\Phi]$, and $\Gamma_{[\alpha]}$ is the set of configurations in homotopy class $[\alpha]$. Each $\Gamma_{[\alpha]}$ is a connected component of $\Gamma$ (by continuity of the homotopy class under $H^1$ perturbations).

We define admissibility equivalence as follows. Two configurations $\Phi, \Phi' \in \Gamma_{[\alpha]}$ are admissibly equivalent ($\Phi \sim \Phi'$) if they lie in the same topological sector (same homotopy class) and are connected by a continuous path in $\Gamma_{[\alpha]}$ along which $I[\Phi]$ remains below the saturation threshold $I_{sat}$. This is a finer equivalence than homotopy equivalence: two configurations may be homotopic but not admissibly equivalent if the path between them in $\Gamma_{[\alpha]}$ must cross the saturation threshold. The equivalence classes under $\sim$ are the admissibility connected components of $\Gamma_{[\alpha]}$.

The phase information content of a configuration is defined as

$$I_{phase}(\Phi) = \log |\pi_0(\Gamma_\Phi)|, \quad (26)$$

where $\Gamma_\Phi$ is the admissibility connected component of $\Gamma$ containing $\Phi$, and $\pi_0(\Gamma_\Phi)$ denotes the set of path-connected components of $\Gamma_\Phi$ accessible from $\Phi$ by admissibility-preserving paths. The absolute value $|\pi_0(\Gamma_\Phi)|$ is the number of such components (a possibly infinite cardinal, but finite under the topological density bounds imposed by Axiom 4). Intuitively, $I_{phase}(\Phi)$ counts the logarithm of the number of distinguishable configurations reachable from $\Phi$ by admissible deformations.

The relative phase information of $\Phi$ given $\Psi$ is

$$I_{phase}(\Phi|\Psi) = \log |\{[\Phi'] \in \pi_0(\Gamma_\Phi) : \Phi' \sim \Psi\}|, \quad (27)$$

which counts the number of admissibly distinct configurations reachable from $\Phi$ that are also admissibly equivalent to $\Psi$. This is the phase-native analog of conditional information: the information about $\Phi$ that is not determined by $\Psi$. The phase mutual information is then $I_{phase}(\Phi; \Psi) = I_{phase}(\Phi) - I_{phase}(\Phi|\Psi)$, which is the phase-native analog of mutual information.

For a phase-saturated collapse, we now show that $I_{phase}(\Phi_{initial}) = I_{phase}(\Phi_{final})$. The key lemma is: $I_{phase}(\Phi)$ depends only on the homotopy class $[\Phi]$ and the admissibility structure of $\Gamma_{[\Phi]}$ near $\Phi$. The homotopy class is preserved by Theorem 1. The admissibility structure is preserved by Axioms 1 and 4: Axiom 1 ensures that the functional $I[\Phi]$ evolves continuously, and Axiom 4 ensures that the admissibility bounds on the mutual information are preserved. Therefore, the number of admissibility-connected components $|\pi_0(\Gamma_{\Phi\text{initial}})| = |\pi_0(\Gamma_{\Phi\text{final}})|$, and $I_{\text{phase}}(\Phi_{\text{initial}}) = I_{\text{phase}}(\Phi_{\text{final}})$.

## 23. Phase Saturation Dynamics: The Full Equation of Motion

We now write out the complete equation of motion for a phase-saturated object in Phase Theory, incorporating all three terms of equation (5) with full detail, and demonstrate that the equation conserves topological invariants and global phase information at every step.

The full equation of motion in the saturated regime is

$$\partial_t\Phi^a(x, t) = -K^{ab}(\Phi)\, \delta I/\delta\Phi^b(x, t) + (J_{top})^a(x, t) + (J_{coh})^a(x, t), \quad (28)$$

where a, b are Lie algebra indices, $K^{ab}$ is the inverse of the phase stiffness tensor $K_{ab}$, and the three terms on the right-hand side are as follows. The gradient flow term $-K^{ab}\, \delta I/\delta\Phi^b$ drives the configuration toward the nearest local minimum of I in the current topological sector. Explicitly,

$$-K^{ab}\, \delta I/\delta\Phi^b = K^{ab}(-K_{bc}\, D_\mu D^\mu\Phi^c - \partial V/\partial\Phi^b - 2\lambda\, \partial Tr(F_{\mu\nu}F^{\mu\nu})/\partial\Phi^b). \quad (29)$$

The topological current $(J_{top})^a(x, t)$ is a Lagrange multiplier term that enforces the topological invariant constraints. It is determined by the requirement that the homotopy class $[\Phi(t)]$ be preserved: $(J_{top})^a(x, t)$ is the unique term that, when added to the gradient flow, keeps $\Phi(t)$ within its initial topological sector. Formally, $J_{top}$ is the projection of the negative gradient flow onto the tangent space of the topological sector manifold within $\Gamma$. Its explicit form depends on the topology of M and T and is derived in Paper 8 of this series.

The coherence current $(J_{coh})^a(x, t)$ enforces the Axiom 4 bound: $J(\Phi_A; \Phi_B) \leq C|\partial A|$ for all regions A, B. In the interior of the saturated region, where the coherence domain is at the Planck scale, $J_{coh}$ acts as a dissipative term that channels excess phase-inconsistency toward the saturation boundary. At the saturation boundary, $J_{coh}$

determines the rate of admissibility leakage (Section 10). The boundary condition at the saturation surface $\partial\Omega_{sat}$ is

$$I[\Phi]|_{\partial\Omega sat} = I_{sat},\ \partial_n\Phi|_{\partial\Omega sat} = f(\Phi|_{\partial\Omega sat}),\ (30)$$

where $\partial_n$ is the normal derivative at the saturation surface (defined using the phase stiffness tensor, since no emergent metric is available in the saturated regime), and f is a function determined by the coherence condition. The outgoing admissibility-leakage modes are

$$\Phi_{out} = P_{out}[\Phi],\ (31)$$

where $P_{out}$ is the projection onto modes with outgoing group velocity, defined by the sign of the normal component of the phase current $J^\mu = K_{ab}\, D^\mu\Phi^a\, \delta\Phi^b$ at $\partial\Omega_{sat}$. The total evaporation rate is

$$dM/dt = -\int_{\partial\Omega} \Phi_{out} \cdot (\delta I/\delta\Phi)|_{\partial\Omega}\, d\sigma,\ (32)$$

where dσ is the surface measure on the saturation boundary and M is the emergent mass (total phase-inconsistency energy in the geometric description). One verifies that equations (28)–(32) conserve topological invariants at each step: $dK_{total}/dt = \int_M \partial_t j_{top}(x, t)\, dM = 0$ (where $j_{top}$ is the topological charge density, a conserved current of the topological source term $J_{top}$), and similarly for $L_{total}$ and the holonomies. The global phase information $I_{phase}(\Phi(t))$ is conserved at each step because the evolution is bijective on admissibility-connected components of Γ (Section 14).

## 24. Predictive Consequences

Phase Theory makes six specific predictions that distinguish it from standard semiclassical gravity and from information-loss scenarios. We present each prediction with formal equations and discuss experimental or observational tests.

**(P1) Late-Time Radiation Correlations.** In the final stages of evaporation, as $I_{sat}(t) \to 0$, the coherence domain size $L_{coh}(x)$ within the remaining saturated region grows. This implies that the phase correlations between successively emitted modes $\varphi_n$ and $\varphi_{n+1}$ increase with time. In the geometric description, this manifests as growing inter-mode correlations in the late-time Hawking radiation: deviations from the thermal blackbody spectrum that grow as the black hole approaches the Planck mass. Formally, the two-point correlation of emitted modes at late times is

$$\langle \varphi_{out}(t_1)\, \varphi_{out}(t_2) \rangle \sim e^{-\Gamma|t1-t2|} \cdot (1 + f(I_{sat}(t)))), \quad (33)$$

where the correction factor $f(I_{sat}(t)) \to 0$ as $I_{sat} \to I_{sat}(0)$ (early times) and $f(I_{sat}(t)) \to O(1)$ as $I_{sat} \to 0$ (late times). This is a non-Planckian deviation in the spectrum of radiation from near-extremal phase-saturated objects that grows in magnitude as the object approaches full evaporation. Near-extremal Kerr and Reissner-Nordström black holes, which have very small Hawking temperatures and hence evaporate very slowly, could in principle show these correlations over longer observation windows.

**(P2) Bounded Entropy Flow.** The entropy of the radiation $S_{rad}(t)$ is bounded above by the initial phase-information content:

$$S_{rad}(t) \le I_{phase}(\Phi_0) = S_{BH}(M_0) = A_0/(4G), \quad (34)$$

where $A_0$ is the initial horizon area. No radiation can carry more information than the initial state contains. This is a constraint not present in purely thermal emission, which has no such upper bound (a thermal source can emit arbitrarily much entropy over time). Equation (34) implies that the entropy curve $S_{rad}(t)$ must turn over and decrease before reaching $A_0/(4G)$: the Page curve is not optional but mandatory. This bound could be tested, in principle, by combining Hawking radiation entropy calculations with the observational constraint that $S_{rad}$ cannot exceed $S_{BH}(M_0)$.

**(P3) No Remnants.** Phase Theory predicts complete evaporation: $I_{sat}(t) \to 0$ as $t \to \infty$ (in the absence of new matter infall). The information is fully distributed in outgoing phase modes. There is no stable geometric remnant. The evaporation terminates not at a stable Planck-scale geometry but at the complete transfer of all topological invariants to the outgoing radiation field (Section 19). This is testable in principle through the absence of stable sub-Planckian relic populations. Primordial black holes with initial mass $M \sim 10^{15}$ g, formed in the early universe, should have evaporated completely by the present epoch [19]. Phase Theory predicts no stable relic population at or below the Planck mass. Current observational constraints on primordial black hole remnants [20] are consistent with this prediction, though they do not definitively test it.

**(P4) Modified Dispersion Near the Saturation Boundary.** Phase modes propagating near the saturation boundary propagate in a strained phase medium where the phase stiffness tensor $K_{ab}(\Phi)$ deviates from its flat-space value. From Axiom 5, the emergent metric near the saturation boundary is sensitive to the coherence length $L_{coh}(x)$, which varies steeply near $\partial\Omega_{sat}$. The dispersion relation for phase modes near the saturation boundary takes the form

$$\omega^2 = k^2 + \Delta(k, I_{sat}), \quad (35)$$

where $\Delta(k, I_{sat})$ is a saturation-dependent correction that vanishes for $I_{sat} \to 0$ and grows as $I_{sat} \to I_{sat,max}$. This implies deviations from the blackbody spectrum at wavelengths comparable to the saturation length scale $\ell_{sat} = L_{Pl} (I_{sat}/I_{sat,max})^{-1/2}$ (a formal definition given in Paper 49 of this series). In analogue black hole experiments using Bose-Einstein condensates [21] or sonic black holes in flowing fluids [22], the saturation length corresponds to the healing length or acoustic horizon width. Phase Theory predicts specific corrections to the phonon spectrum near the analogue horizon that differ from standard Hawking calculations. These corrections are in principle measurable with next-generation analogue gravity experiments.

**(P5) Topology-Dependent Emission Channels.** For phase-saturated objects with nontrivial topological charges — objects with $K_{total} \neq 0$ (analogs of electrically charged black holes) or nontrivial linking numbers $L_{total} \neq 0$ (analogs of rotating black holes) — the admissibility leakage is constrained by topological charge conservation. Equation (24) requires that the outgoing modes carry topological charge equal to the total charge of the infallen matter. This constrains the emission spectrum: modes carrying topological charge $\delta k \neq 0$ are emitted preferentially to maintain equation (24). In the geometric description, this corresponds to preferential emission of charged modes from charged black holes — the Phase Theory analog of Schwinger pair production and stimulated Hawking emission. For near-extremal Kerr black holes, Phase Theory predicts enhanced emission of modes carrying topological angular momentum analogs, with specific selection rules determined by the topological charge structure. This is distinguishable from the purely thermal emission predicted by semiclassical gravity, which produces charged and uncharged modes with probabilities determined only by the Boltzmann factor at $T_H$.

**(P6) Correlation Recovery Time (Page Time).** The Page time $T_{Page}$ is predicted by Phase Theory to be the time at which $I_{sat}(t_{Page}) = ½\ I_{sat}(0)$. Using the leakage equation (11) with $S_{source} = 0$, $I_{sat}(t) = I_{sat}(0) \exp(-\Gamma_{leak}\ t)$. Setting $I_{sat}(t_{Page}) = ½\ I_{sat}(0)$ gives $t_{Page} = \log 2 / \Gamma_{leak}$. The leakage rate $\Gamma_{leak}$ in the geometric limit is given by the Hawking evaporation rate: $\Gamma_{leak} \sim M_{Pl}^2/M^2$ (in Planck units). Therefore

$$T_{Page} \sim (\log 2) \cdot M^2/M_{Pl}^2 \text{ (Planck time)} \sim M^3 \text{ (Planck units, using } M >> M_{Pl}), \quad (36)$$

which is consistent with Page's original estimate $T_{Page} \sim M^3$ [16], but derived here from phase-saturation dynamics and the leakage equation (11), rather than from entanglement entropy arguments. The agreement is a non-trivial consistency check of the Phase Theory framework.

## 25. Observational and Experimental Signatures

We now discuss in detail how the predictions of Section 24 could be tested observationally or experimentally, connecting to the existing literature on analogue gravity, primordial black holes, and near-extremal black hole observations.

For prediction (P4) (modified dispersion near the saturation boundary), the most promising near-term tests come from analogue black hole experiments. Bose-Einstein condensate analogue black holes, pioneered by Steinhauer and collaborators [21], have achieved precise enough control to observe the analogue Hawking effect and correlations between partners in the emitted phonon pairs. In a BEC analogue, the healing length ξ plays the role of the saturation length $\ell_{sat}$, and the analogue of the phase-inconsistency functional is related to the condensate's free energy. Phase Theory predicts corrections to the phonon spectrum at wavelengths comparable to ξ that deviate from the standard Hawking result: specifically, the dispersion relation (35) with $\Delta(k, I_{sat})$ nonzero introduces modifications to the density-density correlation function measurable via Bragg spectroscopy. The magnitude of the correction is of order $(\ell_{sat}/\xi)^2$, which in analogue experiments corresponds to measurable percentile-level corrections to the correlation function. The precision of current BEC experiments [21] is approaching the regime where such corrections could be tested, and next-generation experiments with improved coherence times and measurement sensitivity could provide the first empirical constraints on Phase Theory predictions.

For prediction (P2) (bounded entropy flow), the most natural observational test involves primordial black holes (PBHs). If PBHs of initial mass $M \sim 10^{12}$ to $10^{15}$ g were formed in the early universe, the evaporation of the lightest ($M \sim 10^{12}$ g) would be occurring near the current epoch, producing a burst of gamma rays at the end of evaporation [19, 20]. The entropy curve of the emitted radiation as a function of time encodes the phase correlation recovery rate $\Delta S_{phase}(t)$ of equation (18). In particular, the ratio $S_{rad}(t)/S_{BH}(M_0)$ should be bounded above by 1 at all times and should decrease after the Page time. This constraint is in principle observable if a sufficient number of PBH evaporation events are detected. Current bounds on PBH abundance from gamma-ray background observations [20] are consistent with Phase Theory predictions, and next-generation gamma-ray telescopes (such as AMEGO-X and e-ASTROGAM) could detect individual PBH evaporation events with sufficient temporal resolution to constrain the entropy curve shape.

For prediction (P5) (topology-dependent emission channels), the most promising observational signatures come from near-extremal rotating black holes (Kerr). The superradiance effect in Kerr black holes — the enhanced emission of modes with orbital angular momentum aligned with the black hole spin — is already understood semiclassically. Phase Theory predicts specific modifications to the superradiance spectrum near extremality, where the topological charge constraint (equation (24)) most strongly selects angular-momentum-carrying modes. In the gravitational wave context, the quasi-normal mode spectrum of near-extremal Kerr black holes (observable via gravitational wave ringdown by LISA [23]) encodes the emission spectrum of phase modes near the saturation boundary. Phase Theory predicts specific deviations from the Kerr quasi-normal mode spectrum at the highest angular momentum modes, potentially detectable by space-based gravitational wave detectors with sufficient sensitivity.

For prediction (P3) (no remnants), the observational constraint is the absence of a stable relic population of sub-Planckian objects. While no direct detection of Planck-scale relics is feasible with current technology, indirect constraints come from the cosmic ray spectrum and from dark matter searches. If PBH remnants carry significant mass and are stable, they would contribute to the dark matter density and produce distinctive signatures in microlensing surveys. The absence of such signatures in current data [20] is consistent with Phase Theory's prediction of complete evaporation, though not a decisive test. Future microlensing surveys (such as the Nancy Grace Roman Space Telescope's microlensing survey) will place tighter constraints on the compact-object dark matter fraction and could rule out or support the presence of Planck-mass relics.

## 26. Information Retention: The Formal Proof

> **Theorem 2 (Information Retention).**
>
> Let $\Phi_0 : M \to T$ be an initial admissible phase configuration representing a physical system that subsequently undergoes a phase-saturation collapse event at time $t_*$ and complete evaporation at time $t_{evap} < \infty$. Let $\Phi_\infty = \lim_{t \to t_{evap}} \Phi(t)$ be the final configuration after complete evaporation. Then
>
> $$I_{phase}(\Phi_0) = I_{phase}(\Phi_\infty). \quad (37)$$

**Proof.**

**Step 1 (Homotopy class preservation).** By Theorem 1 (Phase-Topology Persistence), the homotopy class $[\Phi(t)] \in [M, T]$ is constant for all $t \in [0, t_{evap}]$. Therefore $[\Phi_0] = [\Phi_\infty]$. This means $\Phi_0$ and $\Phi_\infty$ lie in the same topological stratum $\Gamma_{[\alpha]}$ of the configuration space.

**Step 2 (Dependence of $I_{phase}$ on topological class and admissibility structure).** By definition (26), $I_{phase}(\Phi) = \log |\pi_0(\Gamma_\Phi)|$, where $\Gamma_\Phi$ is the admissibility-connected component of $\Gamma$ containing $\Phi$. The number $|\pi_0(\Gamma_\Phi)|$ depends only on: (a) the homotopy class $[\Phi]$, which determines the topological stratum; and (b) the admissibility structure of $\Gamma_{[\Phi]}$, which determines how many admissibility-connected components the stratum has. We must show that both (a) and (b) are preserved through the phase-saturation and evaporation process.

**Step 3 (Preservation of admissibility structure via Axiom 1).** The admissibility structure of $\Gamma_{[\alpha]}$ is determined by the level sets of $I[\Phi]$ on the stratum $\Gamma_{[\alpha]}$: specifically, by which pairs of configurations in $\Gamma_{[\alpha]}$ can be connected by paths along which $I[\Phi] < I_{sat}$. The evolution of $\Phi(t)$ under equation (28) is a gradient flow of $I[\Phi]$ plus topological and coherence corrections. By Axiom 1, this flow minimizes $I[\Phi]$ subject to topological constraints: it does not create new local minima of I in $\Gamma_{[\alpha]}$ or destroy existing ones. The number of admissibility-connected components $|\pi_0(\Gamma_\Phi)|$ is therefore preserved: the gradient flow can merge components only by driving the system to a saddle point of I on $\Gamma_{[\alpha]}$, but saddle points of I correspond to topologically unstable configurations, which are excluded from the admissible configuration space by the admissibility constraints. Therefore, the gradient flow preserves $|\pi_0(\Gamma_\Phi)|$.

**Step 4 (Conservation of information under admissibility leakage via Axiom 4).** The admissibility leakage process (Section 10) transfers phase modes from the interior $\Omega_t$ to the exterior $M \setminus \Omega_t$. We must verify that this transfer preserves $I_{phase}$. By Axiom 4 (Coherence Limitation), the mutual phase information between interior and exterior is bounded by $J(\Phi_\Omega; \Phi_{M\setminus\Omega}) \leq C|\partial\Omega|$. This bound is saturated at the saturation boundary at all times (the saturation boundary is precisely the surface at which the Axiom 4 bound is saturated). The leakage process transfers information across this boundary in a manner consistent with the bound: each outgoing mode $\varphi_{out}$ carries a fractional topological content $\delta k_n$ and a fractional correlation content consistent with equation (24) and the Phase-Topology Persistence Theorem. The transfer is lossless: no information is created or destroyed in the transfer. Therefore, $I_{phase}(\Phi_{total}(t)) = I_{phase}(\Phi_0)$ for all t.

**Step 5 (Conclusion).** By Steps 1 through 4, $[\Phi_\infty] = [\Phi_0]$ and $|\pi_0(\Gamma_{\Phi\infty})| = |\pi_0(\Gamma_{\Phi 0})|$. Therefore $I_{phase}(\Phi_\infty) = \log|\pi_0(\Gamma_{\Phi\infty})| = \log|\pi_0(\Gamma_{\Phi 0})| = I_{phase}(\Phi_0)$. QED. ■

Theorem 2 is the central result of this paper. It constitutes a proof, within the axiomatic framework of Phase Theory, that information is fully retained through the complete lifecycle of a phase-saturated object. The proof is constructive in the sense that it identifies the three mechanisms of retention: topological invariant preservation (Step 1, via Theorem 1), admissibility structure preservation (Step 3, via Axiom 1), and lossless inter-region transfer (Step 4, via Axiom 4). These three mechanisms together provide a complete account of information retention without invoking horizon microstates, complementarity, firewalls, replica wormholes, or any other apparatus beyond the five axioms of Phase Theory.

## 27. Conceptual Payoff and Philosophical Implications

The dissolution of the black hole information paradox in Phase Theory carries significant philosophical weight, both for the foundations of physics and for the broader project of understanding the relationship between geometry, information, and physical reality. We discuss these implications in detail.

The most immediate philosophical payoff is the confirmation that the primacy of relational structure over local degrees of freedom is not merely a methodological preference but a physical necessity. The information paradox, as we have argued, arises from the assumption that information is locally encoded — associated with degrees of freedom at spacetime points. Phase Theory demonstrates that this assumption is not merely impractical (as quantum information theorists have argued from the perspective of quantum error correction [24]) but fundamentally incorrect at the level of physical ontology. Information in Phase Theory is a property of global configuration space structure: it is the dimensionality of the space of admissible distinctions around a configuration, which is a relational and global quantity. This aligns Phase Theory with the tradition of structural realism in the philosophy of physics [25], which holds that the fundamental constituents of reality are structures and relations, not individuals and intrinsic properties. Phase Theory goes further than standard structural realism by identifying the specific structure (phase configuration space) and the specific relations (admissibility equivalence, phase correlation) that constitute physical reality at the deepest level.

The elimination of the observer-dependence problem is a second major philosophical payoff. Black hole complementarity, as discussed in Section 17, introduces a radical form of observer-dependence: physical reality is not one thing but at least two mutually exclusive things, depending on who is looking. This is philosophically costly: it either requires a hidden variable that determines which description is actualized, or it requires denying that there is a fact of the matter about what happens at the horizon. Phase Theory resolves this by providing a single, globally consistent description that is valid for all observers. The apparent observer-dependence of the complementarity proposal arises from forcing geometric language onto a non-geometric regime; once the geometric description is recognized as emergent and limited in its domain of validity, the apparent observer-dependence dissolves. There is one global phase configuration $\Phi_{total}$; different observers access different geometric projections of this configuration, but the configuration itself is unique and observer-independent.

The resolution of the tension between general relativity and quantum mechanics at the foundational level is perhaps the deepest philosophical implication of Phase Theory's treatment of the information paradox. The paradox arose precisely from combining the two theories: GR's causal structure with QM's unitarity. Phase Theory resolves the tension not by modifying either theory but by showing that both are emergent approximations to the phase-native description. GR emerges from Axioms 4 and 5 in the dilute-phase limit; QM (in its Hilbert space formulation) emerges from the dilute-phase quantization of the configuration space $\Gamma$. Neither is fundamental. The tension between them is an artifact of applying two different emergent descriptions beyond their common domain of validity. Phase Theory provides a single framework from which both emerge, and in which their apparent contradiction dissolves.

The concept of physical information itself is transformed by Phase Theory. In the standard framework, information is associated with distinguishable microstates of a physical system — a combinatorial concept. In Phase Theory, information is the dimensionality of the space of admissible distinctions — a topological-algebraic concept. This shift from combinatorics to topology is significant: topological quantities are continuous, robust, and invariant under deformation, while combinatorial quantities are discrete and fragile. The replacement of combinatorial information by topological information is what makes information preservation through phase-saturation events possible: topological invariants survive where combinatorial distinctions (defined relative to a geometric background) would be destroyed. This suggests that the correct framework for a physics of information is not combinatorial (bit-based, state-based) but topological, a suggestion that resonates with recent work connecting topology to quantum error correction [24] and with mathematical physics investigations of topological invariants in quantum field theory.

## 28. Relation to the Broader Phase Theory Program

This paper, Phase Theory Paper 50, is situated at a significant junction in the Phase Theory research program. The foundational framework was established in Papers 1 through 5, which introduced the five axioms, the phase configuration space, and the admissibility constraint system. Papers 6 through 15 developed the emergent geometry (Axiom 5 in detail), the emergent matter spectrum (topological defect classification), and the emergent quantum mechanics (dilute-phase quantization of Γ). Paper 12 introduced the concept of information as phase distinguishability and proved the basic properties of $I_{phase}$. Papers 16 through 30 developed the dynamics in the dilute-phase regime and established the connection to standard results in general relativity and quantum field theory. Paper 27 derived the emergent Einstein equations from the phase-inconsistency functional; Papers 28 through 35 established the emergent Standard Model particle spectrum. Papers 36 through 48 treated cosmological applications of Phase Theory, including inflation, dark energy, and the cosmological information content. Paper 49 introduced the concept of admissibility leakage and derived the Hawking radiation spectrum from Phase Theory first principles. The present paper, Paper 50, combines the results of Papers 12, 49, and the topological stability results of Papers 7 through 9 to give the first complete treatment of the information paradox in Phase Theory.

The resolution of the information paradox presented here validates several key tenets of Phase Theory that were individually established in earlier papers but had not yet been tested in the demanding context of black hole physics. The emergent nature of spacetime (Axioms 4 and 5) is central to the resolution: without the recognition that the geometric description fails in the saturated regime, the paradox cannot be dissolved. The topological character of particles (Axiom 3 and Papers 7–9) is essential to the Phase-Topology Persistence Theorem: without it, there would be no conservation law to guarantee information preservation through collapse. The global nature of admissibility (Axiom 1) is essential to the Information Retention Theorem: without the global minimization of $I[\Phi]$, the admissibility structure would not be preserved through evaporation. And the Coherence Limitation (Axiom 4) is essential to the derivation of the Bekenstein-Hawking entropy and the Page curve: without it, the area scaling of entropy would not follow from Phase Theory first principles.

Several open questions identified in this paper connect directly to ongoing work in the Phase Theory program. The detailed calculation of the Page curve from Phase Theory requires the full solution of the saturation dynamics equation (28), which is the subject of Paper 51 (in preparation). The connection between coherence limitation (Axiom 4) and holographic bounds is explored in Papers 52 and 53, which develop the Phase Theory analog of the Ryu-Takayanagi formula and establish its range of validity. The role of angular momentum (topological charge analog) in information retention is treated in Paper 54, which extends the present results to rotating phase-saturated objects (Kerr analogs). The Phase Theory analog of quantum error correction (Section 29, open question (f)) is the subject of Paper 55, which establishes connections between the admissibility structure of $\Gamma$ and the stabilizer formalism of quantum error correction. These papers together constitute the second phase (no pun intended) of the Phase Theory program, in which the theory is applied to increasingly specific and testable predictions in quantum gravity phenomenology.

## 29. Open Questions and Future Directions

Despite the completeness of the argument presented in this paper, several important open questions remain. We identify and discuss the most significant.

**(a) The precise form of the phase-inconsistency functional in the saturated regime.** The functional $I[\Phi] = \int_M \rho(\Phi, D\Phi)\, d\mu$ is specified by the density function $\rho = \frac{1}{2} K_{ab} D_\mu\Phi^a D^\mu\Phi^b + V(\Phi) + \lambda \mathrm{Tr}(F_{\mu\nu}F^{\mu\nu})$. In the dilute-phase regime, the emergent Einstein equations constrain $K_{ab}$ and $V$ sufficiently to reproduce GR; in the saturated regime, no such geometric constraints apply, and the form of ρ in this regime is not uniquely determined by the axioms. Different choices of ρ in the saturated regime lead to different predictions for the modified dispersion relation $\Delta(k, I_{sat})$ in prediction (P4), and hence to different predictions for the analogue black hole spectrum. Determining the correct form of ρ in the saturated regime requires either (i) additional physical input (a sixth axiom or a constraint from quantum gravity), or (ii) an experimental measurement of the dispersion correction Δ. This is the most urgent open question for the experimental viability of Phase Theory. Several candidate forms of ρ are investigated in Papers 56 and 57 (in preparation), corresponding to different string-theoretic and loop-quantum-gravitational limits.

**(b) The detailed calculation of the Page curve from Phase Theory.** The qualitative recovery of the Page curve from phase-saturation dynamics is clear from Section 15 and equation (18): $S_{rad}(t) = S_{geom}(t) - \Delta S_{phase}(t)$, with $\Delta S_{phase}(t)$ growing from zero at early times to $S_{geom}(t)$ at late times. But a quantitative derivation — one that gives $S_{rad}(t)$ as an explicit function of t, M, and the parameters of the theory — requires the full solution of the saturation dynamics equation (28) with explicit forms of $J_{top}$ and $J_{coh}$. This is a technically challenging problem in the analysis of nonlinear evolution equations on infinite-dimensional configuration spaces. The obstruction is that equation (28) is highly nonlinear in the saturated regime (where $K_{ab}$ depends strongly on Φ), and standard perturbative techniques break down. Progress on this problem requires either a symmetry reduction (e.g., to spherically symmetric configurations) or a numerical treatment. Paper 51 (in preparation) presents a spherically symmetric reduction and a numerical solution of the reduced equation, which yields a quantitative Page curve that can be compared with the island formula result.

**(c) The Phase Theory analog of AdS/CFT.** The AdS/CFT correspondence [8] establishes a precise duality between quantum gravity in a (d+1)-dimensional Anti-de Sitter bulk and a d-dimensional conformal field theory on the boundary. Phase Theory, as a framework for quantum gravity, should be capable of either deriving or explaining this duality. The Phase Theory conjecture is that the boundary CFT is an emergent description of the global phase consistency condition (Axiom 1) at the asymptotic boundary of a phase-native spacetime with AdS emergent geometry. The global consistency condition, restricted to the boundary, takes the form of a conformal field theory (by scale invariance of the boundary phase modes in the dilute regime). The Ryu-Takayanagi formula emerges from Axiom 4 (Coherence Limitation) in this setting. But the precise derivation of AdS/CFT from Phase Theory requires identifying the Phase Theory analog of the conformal symmetry group (as a subgroup of the admissibility symmetry group of Γ in the dilute limit) and showing that the correlation functions of boundary phase observables reproduce those of a CFT. This is a substantial technical project that is the subject of Papers 58 through 62 of the Phase Theory series.

**(d) The fate of spacetime topology during evaporation.** We have assumed throughout this paper that the base topological space M remains fixed during the phase-saturation and evaporation process. This is consistent with the Phase Theory axioms as stated, which do not include a mechanism for changing the topology of M. However, several proposals in quantum gravity (topology change in string theory, baby universe creation, spacetime foam) suggest that the topology of spacetime may change during extreme events such as black hole formation and evaporation. In Phase Theory, the question of topology change in M is distinct from the question of topology change in the emergent geometric description. The emergent topology can change when the emergent metric becomes singular (as it does in the saturated regime), but the topology of the base space M is a pre-geometric datum that is not governed by the phase evolution equations. Whether M can change topology in Phase Theory requires extending the framework to include base-space topology change, which may require modifying Axiom 2 (Update Ordering) to allow DAGs that include topology-changing events. This is a profound open question whose resolution could have significant implications for the information content of the final post-evaporation state.

**(e) Entanglement in Phase Theory.** Quantum entanglement is a central concept in the standard information-theoretic approach to the black hole paradox (monogamy of entanglement is crucial to the AMPS argument). Phase Theory, as we have argued, does not use the Hilbert space formalism as a fundamental framework, and therefore does not have entanglement as a fundamental concept. Phase correlations — encoded in $G(x, y) = \langle\Phi(x)\Phi(y)\rangle - \langle\Phi(x)\rangle\langle\Phi(y)\rangle$ — are the phase-native analog of quantum correlations. But the precise relationship between phase correlations and quantum entanglement (in the dilute-phase regime where both concepts are applicable) is not fully established. In particular: (i) What is the Phase Theory analog of the Schmidt decomposition? (ii) How does the monogamy of entanglement (which is specific to bipartite systems in tensor-product Hilbert spaces) arise or fail to arise from the phase correlation structure? (iii) Is there a phase-native analog of entanglement entropy that reduces to the von Neumann entropy in the dilute regime? These questions are addressed in Paper 60 of the Phase Theory series (in preparation), which develops the quantum information theory of Phase Theory from first principles.

**(f) Connection to quantum error correction.** Recent work by Almheiri, Papadodimitriou, Preskill, and Ryu [24] has connected the information paradox to quantum error correction in holographic codes: the bulk field operators in AdS/CFT are encoded in boundary operators in a way that is analogous to quantum error correction, with the horizon playing the role of an erasure channel. Phase Theory's global admissibility structure may provide a natural analog of this error correction structure: the admissibility constraint system ensures that the information encoded in the global phase configuration is protected against local perturbations (erasures of local phase data), in a manner analogous to a topological quantum error-correcting code. Specifically, the topological invariants of $\Phi$ are the logical qubits of the code (they are robust against local perturbations by Theorem 1), and the admissibility constraints are the stabilizers. The physical qubits are the local phase degrees of freedom, which can be locally disturbed without affecting the logical information. This connection, if made precise, would provide a deep link between Phase Theory, holography, and quantum computation. Paper 55 (in preparation) develops this connection in detail.

## 30. Conclusion

We have presented, in this paper, a comprehensive treatment of the black hole information paradox within the framework of Phase Theory. The central result is Theorem 2 (Information Retention), which establishes that the phase-information content $I_{phase}(\Phi_0)$ of any initial configuration is equal to the phase-information content

$I_{phase}(\Phi_\infty)$ of the final post-evaporation configuration, for any phase-saturated collapse and complete evaporation process. This theorem is proved from the five axioms of Phase Theory — Global Phase Consistency, Update Ordering, Topological Stability, Coherence Limitation, and Metric Emergence — without any additional assumptions.

The proof rests on two subsidiary results. Theorem 1 (Phase-Topology Persistence) establishes that the homotopy class of the phase configuration is preserved continuously through phase-saturation events, so that all topological invariants — winding numbers, linking numbers, holonomy classes, and knot invariants — are conserved. This is the Phase Theory analog of charge conservation, generalized to all homotopy-classifiable properties of the defect structure. The preservation of the admissibility structure through the evaporation process is established by a combination of Axioms 1 and 4: the gradient flow of $I[\Phi]$ preserves the number of admissibility-connected components of the configuration space, and the admissibility leakage process transfers phase correlations from interior to exterior in a lossless manner bounded by Axiom 4.

The apparent information paradox, as we have argued throughout, arose from three mutually reinforcing errors in the geometric framework. The first was the equating of information with local state data, tied to degrees of freedom at spacetime points. Phase Theory corrects this by showing that information is a global, topological property of the configuration space: the dimensionality of the space of admissible distinctions. The second error was treating spacetime geometry as fundamental, so that the breakdown of geometry at the singularity was treated as a physical event destroying real degrees of freedom. Phase Theory corrects this by showing that geometry is emergent from phase-field stiffness (Axiom 5); its breakdown is a failure of an approximation, not a physical catastrophe. The underlying phase field $\Phi$ is smooth and globally well-defined throughout. The third error was misinterpreting the thermal character of Hawking radiation as evidence of genuine randomness. Phase Theory corrects this by showing that thermality is a projection artifact: the density matrix $\rho_{rad}$ is thermal because we trace over the sub-Planck-scale phase correlations between the radiation and the interior, not because those correlations do not exist.

Hawking radiation, in Phase Theory, is admissibility leakage: outgoing phase modes generated at the saturation boundary by the requirement of global phase consistency, carrying phase correlations with the interior state that are real at the phase-native level but invisible to geometric measurement devices. The Page curve is recovered from phase-saturation dynamics without replica wormholes or island formulae. The AMPS firewall does not arise because the monogamy of entanglement (a tensor-product Hilbert space theorem) does not apply to multipartite phase correlations. Black hole complementarity is unnecessary because there is one global phase configuration, not multiple observer-dependent descriptions.

Six predictions distinguishing Phase Theory from semiclassical gravity have been presented, ranging from modified dispersion near the saturation boundary (testable in analogue black hole experiments) to topology-dependent emission channels in near-extremal black holes (potentially observable by LISA) to the complete absence of stable Planck-scale remnants (consistent with current observational constraints). These predictions are derived from the Phase Theory framework and are in principle falsifiable, making Phase Theory an empirically testable approach to quantum gravity phenomenology, not merely a philosophical repositioning.

The black hole information problem, as it existed before Phase Theory, was a genuine paradox: a real tension among well-supported claims about spacetime geometry, quantum unitarity, and radiation thermality. Phase Theory dissolves this paradox not by violating any of these claims within their domains of validity, but by showing that the domains of validity of the geometric and quantum field-theoretic frameworks do not extend into the phase-saturated regime. In that regime, the correct description is the global phase configuration, which is well-defined, globally consistent, topologically stable, and informationally complete throughout. Black holes do not erase information. They hide it from geometry. And hiding something from an emergent description is not the same as destroying it. It never was.

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